

Arithmetic Fuchsian groups with signature $(1; e)$

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§ 1. Introduction.

In the previous papers [17], [18] we determined all arithmetic triangle Fuchsian groups. The purpose of this paper is to determine all arithmetic Fuchsian groups with signature $(1; e)$. In § 2, we prove that for arbitrary non-negative integers g and t there exist finitely many arithmetic Fuchsian groups with signature $(g; e_1, e_2, \dots, e_t)$ up to $SL_2(\mathbf{R})$ -conjugation (Theorem 2.1). In § 3 we deal with arithmetic Fuchsian groups Γ with signature $(1; e)$ (i.e. $g=1$, $t=1$). We give a necessary and sufficient condition for such a group Γ to be arithmetic. More precisely, assume that Γ contains -1_2 . Then Γ has the following presentation:

$$\Gamma = \langle \alpha, \beta, \gamma \mid \alpha\beta\alpha^{-1}\beta^{-1}\gamma = -1_2, \gamma^e = -1_2 \rangle,$$

where α and β are hyperbolic elements of $SL_2(\mathbf{R})$ and γ is an elliptic (resp. a parabolic) element such that $\text{tr}(\gamma) = 2\cos(\pi/e)$. Among such triples (α, β, γ) of generators of Γ we can find a certain fundamental triple $(\alpha_0, \beta_0, \gamma_0)$. Let $x = \text{tr}(\alpha_0)$, $y = \text{tr}(\beta_0)$, $z = \text{tr}(\alpha_0\beta_0)$. Then the condition for Γ to be arithmetic can be expressed in terms of x, y, z . We can also obtain an explicit expression of the quaternion algebra associated with Γ (Theorem 3.4). In § 4 using Theorem 3.4 of § 3 we determine all arithmetic Fuchsian groups with signature $(1; e)$ and list them up (Theorem 4.1). In Fricke-Klein [7] we can find some examples of arithmetic Fuchsian groups with signature $(1; e)$.

§ 2. Arithmetic Fuchsian groups.

We recall the definition of arithmetic Fuchsian groups. Let k be a totally real algebraic number field of degree n . Then we have n distinct $\mathbf{Q}$ -embeddings φ_i ($1 \leq i \leq n$) of k into the real number field $\mathbf{R}$, where φ_1 is the identity. Let A be a quaternion algebra over k which is unramified at the place φ_1 and ramified at all other infinite places φ_i ($2 \leq i \leq n$). Then there exists an $\mathbf{R}$ -isomorphism

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$$(2.1) \quad \rho: A \otimes_{\mathbf{Q}} \mathbf{R} \longrightarrow M_2(\mathbf{R}) + \mathbf{H} + \cdots + \mathbf{H},$$

where $\mathbf{H}$ is the Hamilton quaternion algebra over $\mathbf{R}$. Let ρ_1 (resp. ρ_i , $2 \leq i \leq n$) be the composite of $\rho|_A$ with the projection to $M_2(\mathbf{R})$ (resp. $\mathbf{H}$). Then ρ_1 (resp. ρ_i) is a k -isomorphism of A into $M_2(\mathbf{R})$ (resp. $\mathbf{H}$). ρ_1 is uniquely determined up to $GL_2(\mathbf{R})$ -conjugation. We may assume that $\rho_i|_k = \varphi_i$ ($2 \leq i \leq n$). Let O be an order of A . Put $U^{(1)} = \{\varepsilon \in O \mid n_A(\varepsilon) = 1\}$, where $n_A(\cdot)$ is the reduced norm of A over k . Let $\Gamma^{(1)}(A, O) = \rho_1(U^{(1)})$. Then $\Gamma^{(1)}(A, O)$ is a Fuchsian group of the first kind (i.e. a discrete subgroup of $SL_2(\mathbf{R})$ acting discontinuously on the upper half plane $H = \{z \in \mathbf{C} \mid \text{Im}(z) > 0\}$ such that $\text{vol}(H/\Gamma^{(1)}(A, O)) < \infty$, where $\text{vol}(\cdot)$ is the non-Euclidean volume on H .)

DEFINITION 1. Let Γ be a discrete subgroup of $SL_2(\mathbf{R})$ such that $\text{vol}(H/\Gamma) < \infty$. If Γ is commensurable with some $\Gamma^{(1)}(A, O)$, then Γ is called an *arithmetic Fuchsian group*. We call A the *quaternion algebra associated with Γ* .

Let Γ be a Fuchsian group of the first kind with signature $(g; e_1, e_2, \dots, e_t)$, where $2 \leq e_1 \leq e_2 \leq \dots \leq e_t \leq \infty$. Then Γ is generated by $2g$ hyperbolic elements $\{\alpha_i, \beta_i \mid 1 \leq i \leq g\}$ and t elliptic or parabolic elements $\{\gamma_j \mid 1 \leq j \leq t\}$. The fundamental relations among them are given as follows:

$$(2.2) \quad \begin{cases} \alpha_1 \beta_1 \alpha_1^{-1} \beta_1^{-1} \cdots \alpha_g \beta_g \alpha_g^{-1} \beta_g^{-1} \gamma_1 \cdots \gamma_t = \pm 1_2 \\ \gamma_j^{e_j} = \pm 1_2 \quad (1 \leq j \leq t), \end{cases}$$

where we neglect the relation for $e_j = \infty$.

The integer g is the genus of the compact Riemann surface $(H/\Gamma)^*$ obtained by joining the finite number of cusps to H/Γ . We have the following formula:

$$(2.3) \quad \text{vol}(H/\Gamma) = (2\pi)^{-1} \int_{F(\Gamma)} \frac{dx dy}{y^2} = 2g - 2 + \sum_{j=1}^t (1 - 1/e_j) > 0,$$

where $F(\Gamma)$ denotes a fundamental domain of Γ .

Now we shall prove the following theorem.

THEOREM 2.1. *Let g and t be arbitrary non-negative integers. Then there exist only finitely many arithmetic Fuchsian groups with signature $(g; e_1, e_2, \dots, e_t)$ up to $SL_2(\mathbf{R})$ -conjugation.*

PROOF. In order to prove the above theorem we need several propositions and lemmas. Let Γ be an arithmetic Fuchsian group commensurable with $\Gamma^{(1)}(A, O)$. Then by the results of [16] we see that $k = \mathbf{Q}(\text{tr}(\delta) \mid \delta \in \Gamma^{(2)})$, $\rho_1(A) = k[\Gamma^{(2)}]$, where $\Gamma^{(2)}$ is the subgroup of Γ generated by $\{\delta^2 \mid \delta \in \Gamma\}$. Furthermore, $O_k[\Gamma^{(2)}]$ is an order of $\rho_1(A)$, where O_k is the ring of integers in k . Hence there exists a maximal order O_1 in A such that $\Gamma^{(2)}$ is a subgroup of finite index in $\Gamma^{(1)}(A, O_1)$.

PROPOSITION 2.2. *Let Γ be a Fuchsian group with signature $(g; e_1, e_2, \dots, e_t)$. Then the following assertions hold:*

- (i) If $t=0$, then $[\Gamma \cdot \{\pm 1_2\} : \Gamma^{(2)} \cdot \{\pm 1_2\}] = 2^{2g}$.
 (ii) If $t>0$, then $2^{2g} \leq [\Gamma \cdot \{\pm 1_2\} : \Gamma^{(2)} \cdot \{\pm 1_2\}] \leq 2^{2g+t-1}$.

PROOF OF PROPOSITION 2.2. Firstly consider the case (ii). Since $\Gamma \cdot \{\pm 1_2\} / \Gamma^{(2)} \cdot \{\pm 1_2\}$ is an elementary abelian group of type $(2, \dots, 2)$ generated by $2g+t-1$ elements, we see that the second inequality holds. For an arbitrary element γ of Γ we have the expression $\gamma = \pm \delta_1^{m_1} \cdots \delta_r^{m_r}$, where $\delta_j \in \{\alpha_1, \beta_1, \dots, \alpha_g, \beta_g, \gamma_1, \dots, \gamma_t\}$. We put

$$\nu_{\alpha_i}(\gamma) = \sum_{\delta_j = \alpha_i} m_j \pmod{2}, \quad \nu_{\beta_i}(\gamma) = \sum_{\delta_j = \beta_i} m_j \pmod{2}.$$

In view of (2.2), $\nu_{\alpha_i}, \nu_{\beta_i}$ ($1 \leq i \leq g$) are well-defined and they are homomorphisms of Γ onto $\mathbb{Z}/2\mathbb{Z}$. Let $\Gamma_{\alpha_i} = \text{Ker}(\nu_{\alpha_i})$, $\Gamma_{\beta_i} = \text{Ker}(\nu_{\beta_i})$. Then they are pair-wise distinct subgroups of index 2 in Γ . Since $\Gamma^{(2)} \cdot \{\pm 1_2\}$ is contained in $\bigcap_{1 \leq i \leq g} (\Gamma_{\alpha_i} \cap \Gamma_{\beta_i})$, we obtain the first inequality. This proves the assertion (ii). By the same argument we can prove the assertion (i).

Let O_1 be a maximal order of A . Then by a formula of Shimizu [14] we have

$$(2.4) \quad \text{vol}(H/\Gamma^{(1)}(A, O_1)) = 4(2\pi)^{-2n} d(k)^{3/2} \zeta_k(2) \prod_{\mathfrak{p} | D(A)} (n_{k/\mathfrak{q}}(\mathfrak{p}) - 1),$$

where $d(k)$ is the discriminant of k and $\zeta_k(2)$ is the value of the Dedekind zeta function of k at $s=2$ and $D(A)$ is the discriminant of A which is defined by the product of all finite places $\mathfrak{p}$ such that $A \otimes_k k_{\mathfrak{p}}$ is a division quaternion algebra.

Let Γ be an arithmetic Fuchsian group with signature $(g; e_1, \dots, e_t)$ commensurable with $\Gamma^{(1)}(A, O_1)$. Then by (2.3) and (2.4) we have

$$(2.5) \quad 4(2\pi)^{-2n} d(k)^{3/2} \zeta_k(2) \prod_{\mathfrak{p} | D(A)} (n_{k/\mathfrak{q}}(\mathfrak{p}) - 1) = d_1 d_2^{-1} \{2g - 2 + \sum_{1 \leq j \leq t} (1 - 1/e_j)\},$$

where $d_1 = [\Gamma \cdot \{\pm 1_2\} : \Gamma^{(2)} \cdot \{\pm 1_2\}]$, $d_2 = [\Gamma^{(1)}(A, O_1) : \Gamma^{(2)} \cdot \{\pm 1_2\}]$. Since, $\zeta_k(2) > 1$, $\prod_{\mathfrak{p} | D(A)} (n_{k/\mathfrak{q}}(\mathfrak{p}) - 1) \geq 1$, by Proposition 2.2 we have

$$(2.6) \quad d(k) < (2\pi)^{4n/3} \cdot \{2^{2g+t-2}(2g+t-2)\}^{2/3}.$$

On the other hand the following result is proved by A. Odlyzko [11].

PROPOSITION 2.3 (A. Odlyzko). *Let k be a totally real algebraic number field of degree n and $d(k)$ be its discriminant. Then the following inequality holds:*

$$(2.7) \quad d(k) > a^n \exp(-b), \quad \text{where } a = 29.099, b = 8.3185.$$

REMARK. By using a computer he has made a table of the numerical values for a and b . We note that (2.7) is one of them.

If we fix the integers g and t , then by (2.6) and (2.7) we obtain an upper bound of the degree n of k and it is given by

$$(2.8) \quad n_0 = (b + \log_e C(g, t)) / \log_e(a / (2\pi)^{4/3}),$$

where $C(g, t) = 2^{2g+t-2}(2g+t-2)^{2/3}$ and a and b are given in (2.7). We note that $\log_e(a/(2\pi)^{4/3}) = 0.920201\cdots$. Now we fix g , t and n . Then by (2.6) $d(k)$ is bounded. It is well-known that there exist only finitely many algebraic number fields k of given degree such that $d(k)$ is bounded up to $\mathbf{Q}$ -isomorphisms.

Now we may fix the field k . By (2.5) $\prod_{\mathfrak{p} \mid D(A)} (n_{k/\mathbf{Q}}(\mathfrak{p}) - 1)$ is bounded. Therefore, if $\mathfrak{p}$ divides $D(A)$, then $n_{k/\mathbf{Q}}(\mathfrak{p})$ is bounded. Hence there exist only finitely many prime ideals $\mathfrak{p}$ dividing $D(A)$. Thus we have proved that $D(A)$ is of finite possibility. Since A satisfies (2.1), by the Hasse's principle in the theory of simple algebras we see that there exist only finitely many quaternion algebras over k associated with some arithmetic Fuchsian groups with given signature.

We may fix a quaternion algebra A . It is well-known that the type number of maximal orders in A (i.e. the number of conjugate classes of maximal orders under the invertible elements of A) is finite. Hence there exist only finitely many $\Gamma^{(1)}(A, O_1)$ up to $SL_2(\mathbf{R})$ -conjugation. Now by (2.5) we see that d_2 is bounded. We need the following lemma.

LEMMA 2.4. *Let G be a finitely generated group. Then for an arbitrary positive integer d there exist only finitely many subgroups H of G such that $[G : H] \leq d$.*

PROOF OF LEMMA 2.4. We see easily that we may assume that G is a free group. In this case this is a well-known fact (cf. Theorem 7.2.9 p 105 Hall [5]). Q.E.D.

By Lemma 2.4 we see that $\Gamma^{(2)} \cdot \{\pm 1_2\}$ is of finite possibility up to $SL_2(\mathbf{R})$ -conjugation. Let $N(\Gamma^{(2)})$ be the normalizer of $\Gamma^{(2)}$ in $SL_2(\mathbf{R})$. Then we see that $\Gamma \cdot \{\pm 1_2\} \subset N(\Gamma^{(2)})$. We need the following

PROPOSITION 2.5. *Let Γ be a discrete subgroup of $SL_2(\mathbf{R})$ such that $\text{vol}(H/\Gamma) < \infty$. Then the normalizer $N(\Gamma)$ of Γ in $SL_2(\mathbf{R})$ is also a discrete subgroup of $SL_2(\mathbf{R})$ such that $\text{vol}(H/N(\Gamma)) < \infty$ and $[N(\Gamma) : \Gamma] < \infty$.*

The fact that $N(\Gamma)$ is discrete in $SL_2(\mathbf{R})$ is proved in [3] p. 5. Since we have $\text{vol}(H/\Gamma) = [N(\Gamma) : \Gamma \cdot \{\pm 1_2\}] \cdot \text{vol}(H/N(\Gamma))$, we see that the assertion holds. Q.E.D.

By proposition 2.5 we see that there exist only finitely many $\Gamma \cdot \{\pm 1_2\}$ up to $SL_2(\mathbf{R})$ -conjugation. This is valid for Γ . This proves Theorem 2.1.

§ 3. Arithmetic Fuchsian groups with signature $(1; e)$.

From now on we treat Fuchsian groups Γ with signature $(1; e)$ (i.e. $g=1$, $t=1$). Since there is no essential difference between Γ and $\Gamma \cdot \{\pm 1_2\}$, we always assume that Γ contains -1_2 . Then by Fricke-Klein [7] Γ has the following presentation:

$$(3.1) \quad \Gamma = \langle \alpha, \beta, \gamma \mid \alpha\beta\alpha^{-1}\beta^{-1}\gamma = -1_2, \gamma^e = -1_2, \operatorname{tr}(\gamma) = 2 \cos(\pi/e) \rangle,$$

where α, β are hyperbolic elements.

PROPOSITION 3.1. *Let Γ be a Fuchsian group with signature $(1; e)$ ($2 \leq e \leq \infty$). Let $\Gamma^{(2)}$ be the subgroup of Γ generated by $\{\delta^2 \mid \delta \in \Gamma\}$. Then the signature of $\Gamma^{(2)}$ is $(1; e, e, e, e)$ and $[\Gamma: \Gamma^{(2)} \cdot \{\pm 1_2\}] = 4$. Furthermore, let (α, β, γ) be a triple of generators of Γ satisfying (3.1). Then $\Gamma^{(2)} \cdot \{\pm 1_2\}$ is generated by $\{\alpha^2, \beta^2, \beta\gamma\beta^{-1}, \beta\alpha\gamma\alpha^{-1}\beta^{-1}, \gamma, \alpha\gamma\alpha^{-1}\}$ and the field $\mathbf{Q}(\operatorname{tr}(\delta) \mid \delta \in \Gamma^{(2)})$ is generated by $\{(\operatorname{tr}(\alpha))^2, (\operatorname{tr}(\beta))^2, \operatorname{tr}(\alpha)\operatorname{tr}(\beta)\operatorname{tr}(\alpha\beta)\}$ over $\mathbf{Q}$.*

PROOF. Let ν_α, ν_β be the same as defined in Proposition 2.2. Let $\Gamma_\alpha = \operatorname{Ker}(\nu_\alpha)$, $\Gamma_\beta = \operatorname{Ker}(\nu_\beta)$. Then we see that $\Gamma^{(2)} \cdot \{\pm 1_2\} = \Gamma_\alpha \cap \Gamma_\beta$. It is easy to see that γ and $\alpha\gamma\alpha^{-1}$ represent all inequivalent conjugate classes of primitive elliptic (or parabolic if $e = \infty$) elements of Γ_α . Since Γ_α is of index 2 in Γ , we see that the signature of Γ_α is $(1; e, e)$. Moreover, we see that Γ_α is generated by $\{\alpha^2, \beta, \gamma, \alpha\gamma\alpha^{-1}\}$. To see this we denote by Γ' the subgroup of Γ generated by $\{\alpha^2, \beta, \gamma, \alpha\gamma\alpha^{-1}\}$. Then we see easily that Γ' is a normal subgroup of Γ such that $[\Gamma: \Gamma'] \leq 2$. Since Γ' is contained in Γ_α , we see that $\Gamma_\alpha = \Gamma'$. Since $\{1_2, \beta\}$ is a complete set of representatives of $\Gamma_\alpha / \Gamma^{(2)} \cdot \{\pm 1_2\}$, by the same argument as above we see that $\{\gamma, \alpha\gamma\alpha^{-1}, \beta\gamma\beta^{-1}, \beta\alpha\gamma\alpha^{-1}\beta^{-1}\}$ represent all inequivalent conjugate classes of primitive elliptic (or parabolic if $e = \infty$) elements of $\Gamma^{(2)} \cdot \{\pm 1_2\}$ and that the signature of $\Gamma^{(2)} \cdot \{\pm 1_2\}$ is $(1; e, e, e, e)$. Let Γ'' be the subgroup of Γ generated by $\{\alpha^2, \beta^2, \gamma, \alpha\gamma\alpha^{-1}, \beta\gamma\beta^{-1}, \beta\alpha\gamma\alpha^{-1}\beta^{-1}\}$. Then we see that $\Gamma'' \subset \Gamma_\alpha \cap \Gamma_\beta = \Gamma^{(2)} \cdot \{\pm 1_2\}$. Using the relations: $\beta\alpha^2\beta^{-1} = \gamma(\alpha\gamma\alpha^{-1})\alpha^2$, $\beta^{-1}\alpha^2\beta = \beta^{-2}(\beta\alpha^2\beta^{-1})\beta^2$, $\beta^{-1}(\alpha\gamma\alpha^{-1})\beta = \beta^{-2}(\beta\alpha\gamma\alpha^{-1}\beta^{-1})\beta^2$, we see that β normalizes Γ'' . By the relation $\alpha\gamma\alpha^{-1} = \gamma^{-1}\beta\alpha^2\beta^{-1}\alpha^{-2}$ we see that Γ_α is generated by $\{\alpha^2, \beta, \gamma\}$. Therefore, Γ'' is a normal subgroup of Γ_α such that $[\Gamma_\alpha: \Gamma''] \leq 2$. Hence we see that $\Gamma'' = \Gamma^{(2)} \cdot \{\pm 1_2\}$. Let $k = \mathbf{Q}(\operatorname{tr}(\alpha^2), \operatorname{tr}(\beta^2), \operatorname{tr}(\alpha^2\beta^2))$. By the equations

$$(3.2) \quad \begin{cases} \operatorname{tr}(\alpha^2) = \operatorname{tr}(\alpha)^2 - 2, \operatorname{tr}(\beta^2) = \operatorname{tr}(\beta)^2 - 2, \\ \operatorname{tr}(\alpha^2\beta^2) = \operatorname{tr}(\alpha)\operatorname{tr}(\beta)\operatorname{tr}(\alpha\beta) - \operatorname{tr}(\alpha)^2 - \operatorname{tr}(\beta)^2 + 2, \end{cases}$$

we see that $k = \mathbf{Q}(\operatorname{tr}(\alpha)^2, \operatorname{tr}(\beta)^2, \operatorname{tr}(\alpha)\operatorname{tr}(\beta)\operatorname{tr}(\alpha\beta))$. Let A be the vector space spanned by $\{1_2, \alpha^2, \beta^2, \alpha^2\beta^2\}$ over k in $M_2(\mathbf{R})$. By the equations $\beta^2\alpha^2 = \operatorname{tr}(\alpha^2\beta^2)1_2 - \alpha^{-2}\beta^{-2}$, $\alpha^{-2} = \operatorname{tr}(\alpha^2)1_2 - \alpha^2$, $\beta^{-2} = \operatorname{tr}(\beta^2)1_2 - \beta^2$, we see that A is an algebra over k . Using the equation $\delta = \operatorname{tr}(\delta)^{-1}(\delta^2 + 1_2)$ for $\delta \in SL_2(\mathbf{R})$ such that $\operatorname{tr}(\delta) \neq 0$, we see that $\gamma = -\beta\alpha\beta^{-1}\alpha^{-1} = \operatorname{tr}(\alpha)^{-2}\operatorname{tr}(\beta)^{-2}(\beta^2 + 1_2)(\alpha^2 + 1_2)(\beta^{-2} + 1_2)(\alpha^{-2} + 1_2) \in A$. In the same way we see that $\alpha\gamma\alpha^{-1}$, $\beta\gamma\beta^{-1}$ and $\alpha\beta\gamma\beta^{-1}\alpha^{-1}$ are also contained in A . It follows that $A = k[\Gamma^{(2)}]$ and $k = \mathbf{Q}(\operatorname{tr}(\delta) \mid \delta \in \Gamma^{(2)})$. Q.E.D.

Let Γ be a Fuchsian group with signature $(1; e)$. Let $\{\alpha, \beta, \gamma\}$ be a triple of generators of Γ satisfying (3.1). Let $x = \operatorname{tr}(\alpha)$, $y = \operatorname{tr}(\beta)$, $z = \operatorname{tr}(\alpha\beta)$. Then by the equation $\operatorname{tr}(\delta\varepsilon) + \operatorname{tr}(\delta\varepsilon^{-1}) = \operatorname{tr}(\delta)\operatorname{tr}(\varepsilon)$ for $\delta, \varepsilon \in SL_2(\mathbf{R})$ and by (3.1) we have the following equation (cf. Fricke-Klein [7] p. 306)

$$(3.3) \quad x^2 + y^2 + z^2 - xyz = 2 - 2 \cos(\pi/e).$$

Now we consider the transformations:

- (i) $\alpha_1 = -\alpha, \beta_1 = -\beta, \gamma_1 = \gamma,$
- (ii) $\alpha_2 = -\alpha, \beta_2 = \beta, \gamma_2 = \gamma,$
- (iii) $\alpha_3 = \alpha, \beta_3 = -\beta, \gamma_3 = \gamma,$
- (iv) $\alpha_4 = \beta, \beta_4 = \alpha, \gamma_4 = \gamma^{-1},$
- (v) $\alpha_5 = \alpha\beta, \beta_5 = \alpha^{-1}, \gamma_5 = \gamma,$
- (vi) $\alpha_6 = \alpha^{-1}, \beta_6 = \alpha\beta\alpha^{-1}, \gamma_6 = \gamma^{-1}.$

Then each $(\alpha_i, \beta_i, \gamma_i)$ ($1 \leq i \leq 6$) is also a triple of generators of Γ satisfying (3.1).

Let $x_i = \text{tr}(\alpha_i)$, $y_i = \text{tr}(\beta_i)$, $z_i = \text{tr}(\alpha_i\beta_i)$. Then (x_i, y_i, z_i) is given by

- (i)' $(x_1, y_1, z_1) = (-x, -y, z),$
- (ii)' $(x_2, y_2, z_2) = (-x, y, -z),$
- (iii)' $(x_3, y_3, z_3) = (x, -y, -z),$
- (iv)' $(x_4, y_4, z_4) = (y, x, z),$
- (v)' $(x_5, y_5, z_5) = (z, x, y),$
- (vi)' $(x_6, y_6, z_6) = (x, y, xy - z).$

We note that each (x_i, y_i, z_i) ($1 \leq i \leq 6$) also satisfies (3.3).

DEFINITION 2. Let notations be the same as above. Each transformation $(\alpha, \beta, \gamma) \rightarrow (\alpha_i, \beta_i, \gamma_i)$ ($1 \leq i \leq 6$) is called an *elementary operation* for (α, β, γ) .

These operations are introduced in Fricke-Klein [7].

DEFINITION 3. Let (α, β, γ) be a triple of generators of Γ satisfying (3.1). We denote the *height* of (α, β, γ) by

$$(3.4) \quad h(\alpha, \beta, \gamma) = \text{tr}(\alpha)^2 + \text{tr}(\beta)^2 + \text{tr}(\alpha\beta)^2.$$

This notion is a modified one given in Mordell [10] p. 107. We note here that each permutation of (x, y, z) can be realized by a finite number of the elementary operations. The height $h(\alpha, \beta, \gamma)$ is unchanged under the operations (i), (ii), (iii), (iv), (v) and by the operation (vi) we have

$$(3.5) \quad h(\alpha_6, \beta_6, \gamma_6) = h(\alpha, \beta, \gamma) + x^2y^2 - 2xyz.$$

DEFINITION 4. Let (α, β, γ) and $(\alpha', \beta', \gamma')$ be arbitrary triples of generators of Γ satisfying (3.1). If the one can be obtained from the other under a finite number of the elementary operations, we say that they are *equivalent to each other* and we denote $(\alpha, \beta, \gamma) \sim (\alpha', \beta', \gamma')$.

This is obviously an equivalence relation.

DEFINITION 5. Let $(\alpha_0, \beta_0, \gamma_0)$ be a triple of generators of Γ satisfying (3.1). We call $(\alpha_0, \beta_0, \gamma_0)$ a *fundamental triple of generators* if it satisfies the following conditions:

$$(3.6) \quad 2 < \text{tr}(\alpha_0) \leq \text{tr}(\beta_0) \leq \text{tr}(\alpha_0\beta_0),$$

$$(3.7) \quad h(\alpha_0, \beta_0, \gamma_0) = \text{Min} \{h(\alpha, \beta, \gamma) \mid (\alpha, \beta, \gamma) \sim (\alpha_0, \beta_0, \gamma_0)\}.$$

This definition is motivated by the notion given in Mordell [10] p. 107.

PROPOSITION 3.2. *Let (α, β, γ) be a triple of generators of Γ satisfying (3.1). Then by a finite number of the elementary operations (α, β, γ) can be transformed to a fundamental triple of generators of Γ .*

PROOF. Let $h=h(\alpha, \beta, \gamma)$. Let C_h be the set of all triples $(\alpha', \beta', \gamma')$ such that $(\alpha', \beta', \gamma') \sim (\alpha, \beta, \gamma)$ and $h(\alpha', \beta', \gamma') \leq h$. Then we have $|\text{tr}(\alpha')| \leq h^{1/2}$, $|\text{tr}(\beta')| \leq h^{1/2}$. By a result of [3] p. 88 (and Takeuchi [15]) the set $\text{tr}(\Gamma)$ has no limit point in $\mathbf{R}$. Hence C_h is a finite set. Therefore, we can find a triple $(\alpha_0, \beta_0, \gamma_0)$ equivalent to (α, β, γ) satisfying (3.7). Now we need the following

LEMMA 3.3. *Let Γ be a Fuchsian group with signature $(1; e)$. Let (α, β, γ) be a triple of generators of Γ satisfying (3.1). Then $\alpha\beta$ is a hyperbolic element and $\text{tr}(\alpha)\text{tr}(\beta)\text{tr}(\alpha\beta) \geq 10$.*

PROOF. Assume that $\alpha\beta$ is non-hyperbolic. Then we have the expression $\alpha\beta = \pm \delta^{-1} \gamma^m \delta$ for $\delta \in \Gamma$. Since $\nu_\alpha(\alpha\beta) = 1$ and $\nu_\alpha(\delta^{-1} \gamma^m \delta) = \nu_\alpha(\gamma^m) = 0$, we have a contradiction. This shows $\alpha\beta$ is a hyperbolic element. Since $|\text{tr}(\alpha)| > 2$, $|\text{tr}(\beta)| > 2$, $|\text{tr}(\alpha\beta)| > 2$, by (3.3) we have $\text{tr}(\alpha)\text{tr}(\beta)\text{tr}(\alpha\beta) > 10 + 2 \cos(\pi/e) \geq 10$. This proves Lemma 3.3.

By Lemma 3.3 under a finite number of operations (i)–(v) we obtain a triple of generators of Γ satisfying (3.6) and (3.7). This proves Proposition 3.2. Q.E.D.

In order to determine all arithmetic Fuchsian groups with signature $(1; e)$ we shall prove the following theorem.

THEOREM 3.4. *Let Γ be an arithmetic Fuchsian groups with signature $(1; e)$. Let A be the quaternion algebra over k associated with Γ . Assume that Γ contains -1_2 . Let (α, β, γ) be a fundamental triple of generators of Γ satisfying (3.1). Put*

$$(3.8) \quad x = \text{tr}(\alpha), \quad y = \text{tr}(\beta), \quad z = \text{tr}(\alpha\beta).$$

Then the following assertions hold:

- (i) $k = \mathbf{Q}(x^2, y^2, z^2, xyz)$ and k contains $\cos(\pi/e)$.
- (ii) x, y and z are algebraic integers satisfying (3.9), (3.10), (3.11):

$$(3.9) \quad x^2 + y^2 + z^2 - xyz = c_e, \text{ where } c_e = 2 - 2 \cos(\pi/e) \text{ } (c_e = 0 \text{ if } e = \infty).$$

$$(3.10) \quad \begin{cases} 2 < x < 3 \text{ } (2 < x \leq 3 \text{ if } e = \infty), \\ 4(x^2 - c_e)/(x^2 - 4) \leq y^2 \leq (x^2 - c_e)/(x - 2), \\ x \leq y \leq z = (xy - \sqrt{x^2 y^2 - 4x^2 - 4y^2 + 4c_e})/2. \end{cases}$$

$$(3.11) \quad \begin{cases} 0 < \varphi_i(y^2) \leq \varphi_i(4(x^2 - c_e)/(x^2 - 4)) < 4, \\ 0 < \varphi_i(z^2) \leq \varphi_i(4(y^2 - c_e)/(y^2 - 4)) < 4, \\ 0 < \varphi_i(x^2) \leq \varphi_i(4(z^2 - c_e)/(z^2 - 4)) < 4 \text{ } (2 \leq i \leq n). \end{cases}$$

(iii) $A \cong \left(\frac{a, b}{k}\right)$, where $a = x^2(x^2 - 4)$, $b = -(2 + 2 \cos(\pi/e))x^2y^2$. We denote by $\left(\frac{a, b}{k}\right)$ a quaternion algebra over k defined as follows:

$$\left(\frac{a, b}{k}\right) = k1_2 + k\omega + k\Omega + k\omega\Omega, \quad \omega^2 = a, \quad \Omega^2 = b, \quad \omega\Omega + \Omega\omega = 0.$$

Conversely, let x, y and z be algebraic integers satisfying (i), (ii). Let α, β be two elements of $SL_2(\mathbf{R})$ determined by (3.8). Then the subgroup of $SL_2(\mathbf{R})$ generated by $\{\alpha, \beta\}$ is an arithmetic Fuchsian group with signature $(1; e)$.

REMARK. By (3.11) in particular we have

$$(3.12) \quad 0 < \varphi_i(x^2), \quad \varphi_i(y^2), \quad \varphi_i(z^2) < \varphi_i(c_e) \quad (2 \leq i \leq n).$$

In case $e = \infty$, this means that $n = 1$. Hence $k = \mathbf{Q}$. In fact $A \cong M_2(\mathbf{Q})$.

PROOF OF THEOREM 3.4. Let Γ be commensurable with $\Gamma^{(1)}(A, O)$. Then there exists a maximal order O_1 of A such that $\Gamma^{(2)}$ is a subgroup of index finite in $\Gamma^{(1)}(A, O_1)$. $k = \mathbf{Q}(\text{tr}(\delta) | \delta \in \Gamma^{(2)})$ and $\text{tr}(\Gamma^{(2)})$ is contained in the ring O_k of integers in k (cf. [16]). Since $\rho_i(A)$ ($2 \leq i \leq n$) is contained in $\mathbf{H}$, we have $\varphi_i(\text{tr}(\alpha^2)) = \text{tr}_{\mathbf{H}/\mathbf{R}}(\rho_i(\alpha^2))$ is contained in the interval $(-2, 2)$. By the equation $x^2 = \text{tr}(\alpha^2) + 2$ we see that x^2 is an algebraic integer in k such that $4 < x^2$, $0 < \varphi_i(x^2) < 4$ ($2 \leq i \leq n$). Hence x is totally real. In the same way we see that y and z are also totally real algebraic integers.

Since (α, β, γ) is a fundamental triple of generators of Γ , we have $h(\alpha, \beta, \gamma) \leq h(\alpha_0, \beta_0, \gamma_0)$. By (3.5) we see that $x \leq y \leq z \leq xy/2$. Hence by (3.3) $x^2y^2 - 4x^2 - 4y^2 + 4c_e \geq 0$ and $z = (xy - \sqrt{x^2y^2 - 4x^2 - 4y^2 + 4c_e})/2$. Let $f(t) = t^2 - xyt$ ($y \leq t \leq xy/2$). Then we see easily that $y^2(1-x) \geq f(t) \geq -x^2y^2/4$. Hence by (3.3) we have the second and third inequality of (3.10). Now we shall prove the first inequality of (3.10). By the inequality $3z^2 - xyz \geq x^2 + y^2 + z^2 - xyz = c_e > 0$ in case $e < \infty$, we have $xy/3 < z \leq xy/2$. Hence $-xy/6 < z - xy/2 \leq 0$. By (3.3) $x^2 + y^2 + (z - xy/2)^2 - x^2y^2/4 = c_e$. Thus we have $2y^2(9 - x^2)/9 \geq x^2 + y^2 - 2x^2y^2/9 > c_e \geq 0$. Hence we have $2 < x < 3$ in case $e < \infty$. In case $e = \infty$ by the slight modification of the above argument we have $2 < x \leq 3$ (cf. Mordell [10] p. 91). Since z is totally real, by (3.3) we have $\varphi_i(x^2y^2 - 4x^2 - 4y^2 + 4c_e) \geq 0$. By the same argument we can prove all inequalities of (3.11).

We shall prove the assertion (iii). By Proposition 3.1 and its proof we see that $k = \mathbf{Q}(\text{tr}(\delta) | \delta \in \Gamma^{(2)}) = \mathbf{Q}(x^2, y^2, xyz)$ and $k \ni c_e$. Let $A_0 = k[\Gamma^{(2)}]$ be the vector space spanned by $\Gamma^{(2)}$ over k in $M_2(\mathbf{R})$. Then $A_0 = k1_2 + k\alpha^2 + k\beta^2 + k\alpha^2\beta^2 = \rho_1(A)$. Let $\xi = y_01_2 + y_1\alpha^2 + y_2\beta^2 + y_3\alpha^2\beta^2$ be an arbitrary element of A_0 ($y_i \in k$). Let $c_1 = \text{tr}(\alpha^2)$, $c_2 = \text{tr}(\beta^2)$, $c_3 = \text{tr}(\alpha^2\beta^2)$, $c_4 = \text{tr}(\alpha^2\beta^{-2})$. Then the reduced norm $n_{A_0}(\xi)$ of ξ is given by

$$n_{A_0}(\xi) = (y_0, y_1, y_2, y_3) D_0^t (y_0, y_1, y_2, y_3),$$

where

$$D_0 = \begin{pmatrix} 1, & c_1/2, & c_2/2, & c_3/2 \\ c_1/2, & 1, & c_4/2, & c_2/2 \\ c_2/2, & c_4/2, & 1, & c_1/2 \\ c_3/2, & c_2/2, & c_1/2, & 1 \end{pmatrix}.$$

By the following linear transformation:

$$\begin{cases} Y_0 = y_0 + (c_1 y_1 + c_2 y_2 + c_3 y_3)/2, \\ Y_1 = y_1/2 - ((c_1 c_2 - 2c_3)y_2 + (c_1 c_3 - 2c_2)y_3)/(2(4 - c_1^2)), \\ Y_2 = y_3/2, \\ Y_3 = (y_2 + c_1 y_3/2)/(4 - c_1^2), \end{cases}$$

we have

$$\begin{aligned} n_{A_0}(\xi) &= Y_0^2 + (4 - c_1^2)Y_1^2 - (c_1^2 + c_2^2 + c_3^2 - c_1 c_2 c_3 - 4)Y_2^2 \\ &\quad - (4 - c_1^2)(c_1^2 + c_2^2 + c_3^2 - c_1 c_2 c_3 - 4)Y_3^2. \end{aligned}$$

Since $c_1 = x^2 - 2$, $c_2 = y^2 - 2$, $c_3 = -x^2 - y^2 + xyz + 2$, by an easy calculation we see that A_0 is isomorphic to $\left(\frac{a, b}{k}\right)$, where a, b are as given in (iii).

Conversely, let x, y, z be algebraic integers satisfying (i), (ii). Let α, β be two elements of $SL_2(\mathbf{R})$ determined by (3.8). Then α, β are uniquely determined up to $GL_2(\mathbf{R})$ -conjugation. We can define γ so that (α, β, γ) satisfies (3.1). Now we need the following proposition proved in Fricke-Klein [7] pp. 335-353 and Purzitsky-Rosenberger [13].

PROPOSITION 3.5. *Let α, β be two elements of $SL_2(\mathbf{R})$ such that $2 < \text{tr}(\alpha)$, $2 < \text{tr}(\beta)$, $\text{tr}(\alpha\beta\alpha^{-1}\beta^{-1}) = -2 \cos(\pi/e)$ ($= -2$ if $e = \infty$). Then the subgroup of $SL_2(\mathbf{R})$ generated by $\{\alpha, \beta\}$ is a Fuchsian group of the first kind with signature $(1; e)$.*

By Proposition 3.5 the subgroup Γ of $SL_2(\mathbf{R})$ generated by $\{\alpha, \beta\}$ is a Fuchsian group with signature $(1; e)$. Let $k = \mathbf{Q}(\text{tr}(\delta) | \delta \in \Gamma^{(2)})$ and $A_0 = k[\Gamma^{(2)}]$. Then by the same argument as before we see that $k = \mathbf{Q}(x^2, y^2, xyz)$ and k contains $\cos(\pi/e)$ and $A_0 = \left(\frac{a, b}{k}\right)$. By (3.12) we see that A_0 is unramified at φ_1 and ramified at all other φ_i ($2 \leq i \leq n$). Since Γ is generated by $\{\alpha, \beta\}$, by Lemma 2 in [17] p. 95 we see that $\text{tr}(\Gamma)$ is contained in the ring of integers in $\mathbf{Q}(x, y, z)$. Let $O = O_k[\Gamma^{(2)}]$ be the O_k -module generated by $\Gamma^{(2)}$ in $M_2(\mathbf{R})$. Then O is an order of A_0 and $\Gamma^{(2)}$ is a subgroup of $\Gamma^{(1)}(A_0, 0)$ of finite index. This shows that Γ is arithmetic. This completes the proof of Theorem 3.4. Q.E.D.

The following theorem is useful to determine all arithmetic Fuchsian groups with signature $(1; e)$.

THEOREM 3.6. *Let k be a totally real algebraic number field of degree n such that k contains $\cos(\pi/e)$ ($2 \leq e < \infty$). Let $c_e = 2 - 2\cos(\pi/e)$. If there exists an algebraic integer X in k satisfying the inequalities:*

$$(3.13) \quad 4 < X < 9, \quad 0 < \varphi_i(X) < \varphi_i(c_e) \quad (2 \leq i \leq n),$$

then (e, n) is one of pairs listed below:

$$(e, n) = (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), \\ (3, 4), (4, 2), (4, 4), (4, 6), (4, 8), (5, 2), (5, 4), (6, 2), (6, 4), \\ (6, 6), (7, 3), (7, 6), (8, 4), (8, 8), (9, 3), (9, 6), (10, 4), (11, 5), \\ (12, 4), (12, 8), (13, 6), (14, 6), (15, 4), (15, 8), (16, 8), (17, 8), \\ (18, 6), (19, 9), (20, 8), (21, 6), (24, 8), (25, 10), (27, 9), (30, 8), \\ (33, 10).$$

PROOF. By (3.13) we have

$$0 < X(X - c_e) < 9(9 - c_e), \quad 0 < \varphi_i(X(c_e - X)) \leq \varphi_i(c_e^2)/4 \quad (2 \leq i \leq n).$$

Since X is an algebraic integer in k , we have

$$(3.14) \quad 1 \leq |n_{k/Q}(X(c_e - X))| < (9/c_e)(9/c_e - 1)n_{k/Q}(c_e^2)/4^{n-1}.$$

Hence we have

$$(3.15) \quad 4^{n-1} < (9/c_e)(9/c_e - 1)n_{k/Q}(c_e^2).$$

Now we need the following

LEMMA 3.7. Let $c_e = 2 - 2\cos(\pi/e)$. Then the following assertions hold:

- (i) If $e \neq 2^m$, then c_e is a unit of the ring of integers in the field $\mathbf{Q}(\cos(\pi/e))$.
- (ii) If $e = 2^m$, then $n_{\mathbf{Q}(\cos(\pi/e))/\mathbf{Q}}(c_e) = 2$.

The proof of this lemma is referred to Lehmer [8] and Liang [9].

Let $k_0 = \mathbf{Q}(\cos(\pi/e))$, $n_1 = [k : k_0]$. Then we have $n = n_1 \cdot \varphi(2e)/2$, where $\varphi(\cdot)$ is the Euler function. We divide into two cases: $e = 2^m$ and $e \neq 2^m$. Firstly consider the case $e \neq 2^m$. By (3.15) and Lemma 3.7 we have

$$(3.16) \quad 2^{\varphi(2e)/2} < 9/(1 - \cos(\pi/e)).$$

Since $t^2/2 - t^4/24 < 1 - \cos(t)$ ($0 < t$), we have

$$(3.17) \quad 2^{\varphi(2e)/2} < 18e^2/(\pi^2(1 - 12^{-1}(\pi/e)^2)).$$

It is known that for an arbitrary $\delta > 0$, $\lim_{m \rightarrow \infty} \varphi(m)/m^{1-\delta} = \infty$ (cf. Hardy-Wright [6] Theorem 3.27). Using this result we can prove that there exist only a finite number of such numbers e . By (3.15) we see that there are also finitely many such numbers n . In order to determine the pair (e, n) more precisely we need the following

LEMMA 3.8. If $43 \leq m$, then $m^{2/3} \leq \varphi(m)$.

PROOF. Let $m = p_1^{e_1} \cdots p_r^{e_r}$ be the prime divisors decomposition, where p_i is a prime number such that $p_1 < p_2 < \cdots < p_r$ and $e_i \geq 1$. Let p be a prime number. If $e \geq 3$, then $p^{e-3}(p-1) \geq 1$. Let $\phi(m) = \varphi(m)^3/m^2$. Then we have $\phi(m) = \prod_{1 \leq i \leq r} p_i^{e_i-3}(p_i-1)^3$. It suffices to prove that $\phi(m) \geq 1$ for $m \geq 43$. By an easy calculation we have $\phi(2) = 1/4$, $\phi(3) = 8/9$, $\phi(2^2) = 1/2$, $\phi(3^2) = 8/3$. Furthermore, for

an arbitrary prime number p such that $p \geq 5$ we see that $\phi(p) = p - 3 + (3 - 1/p)/p > 1$ and $\phi(p^2) = p^2 - 3p + 3 - 1/p > 1$. Therefore, we see easily that if there is a p_i such that $p_i \geq 11$, then $\phi(m) > 1$. Now we may assume that $m = 2^{e_1} 3^{e_2} 5^{e_3} 7^{e_4}$ ($0 \leq e_i$). We distinguish several cases. If $e_4 \geq 2$, then $\phi(m) \geq (1/4)(8/9)(6^3/7^2) > 1$. Consider the case $e_4 = 1$. If $e_1 = 1, e_2 = 1$, then by the condition $43 \leq m$ we have $e_3 \geq 1$. Hence $\phi(m) \geq (1/4)(8/9)(4^3/5^2)(6^3/7^2) > 1$. If $e_1 \geq 2$ or $e_2 \geq 2$, then $\phi(m) \geq (1/2)(8/9)(6^3/7^2) > 1$ or $\geq (1/4)(6^3/7^2) > 1$. We may consider the case $m = 2^{e_1} 3^{e_2} 5^{e_3}$. If $e_3 \geq 2$, then $\phi(m) \geq (1/4)(8/9)(4^3/5) > 1$. If $e_3 = 1$, then by the condition $43 \leq m$ we have $e_1 \geq 2$ or $e_2 \geq 2$. Hence we see easily that $\phi(m) > 1$. It remains the case $m = 2^{e_1} 3^{e_2}$. By the similar argument as above we can verify our assertion. Q.E.D.

Now we return to the proof of Theorem 3.6. We assume that $e \geq 32$. Then by Lemma 3.8 we have $(2e)^{2/3} \leq \varphi(2e)$. Hence by (3.17) and by the inequality $\pi/32 < 1/10$ we have $2^{(2e)^{2/3}/2} < 18(1200/1199)\pi^{-2}e^2$. Let $t = (2e)^{2/3}/2$. Then we have $8 \leq t$ and $2^t < 36(1200/1199)\pi^{-2}t^3$. We note that the approximate value of $36(1200/1199)\pi^{-2}$ is 3.6506. Let $f(t) = 2^t/t^3$. Then $f(t)$ is monotone increasing on $8 \leq t$. Since $f(13) \div 3.7287$, we see that $t < 13$. Hence we have $e \leq 66$. For each $e \leq 66$ such that $e \neq 2^m$ we examine (3.16) and by (3.15) we obtain the pairs listed in Theorem 3.6.

Next let us consider the case $e = 2^m$. Assume that $m \geq 2$. In this case we denote $d = 2^{m-1} = \varphi(2e)/2$. By (3.14) and Lemma 3.7 we have

$$(3.18) \quad 4^{n-n_1-1} < (9/c_e)(9/c_e - 1).$$

By the assumption $2 \leq m$ we have $2 \leq d$. By (3.18) we have $2^{d-1} \leq 2^{(d-1)n_1} < 9/(1 - \cos(\pi/e))$. Hence $2^{e/2} < 36e^2/(\pi^2(1 - (\pi/e)^2/12))$. Assume that $e \geq 32$. Then by the same argument as in the case $e \neq 2^m$ we have $e < 22$. This is a contradiction. Thus we see that $e = 4, 8, 16$. For each $e = 4, 8, 16$ by (3.18) we can determine all n .

Let us consider the case $e = 2$. In this case the above argument does not work. By (3.13) we have $8 < X(X-2)(X-1)^2 < 63 \cdot 64 (= 4032)$, $0 < \varphi_i(X(2-X)(X-1)^2) \leq 1/4$ ($2 \leq i \leq n$). Since X is an algebraic integer, we have $1 \leq |n_{k/Q}(X(2-X)(X-1)^2)|$. Hence we have $4^{n-1} < 4032$. Therefore, we have $n \leq 6$. This completes the proof of Theorem 3.6.

§ 4. Determination of all arithmetic Fuchsian groups with signature $(1; e)$.

4.1. In this section we shall determine explicitly all arithmetic Fuchsian groups Γ with signature $(1; e)$. In order to do this it suffices to give a fundamental triple (α, β, γ) of generators of Γ . Let $x = \text{tr}(\alpha)$, $y = \text{tr}(\beta)$, $z = \text{tr}(\alpha\beta)$. Then (α, β, γ) is uniquely determined by (x, y, z) up to $GL_2(\mathbf{R})$ -conjugation. The conditions for Γ to be arithmetic are given in terms of (x, y, z) in Theorem 3.4

§3. In the following theorem we shall give a complete list of all (x, y, z) such that the group generated by (α, β, γ) obtained from (x, y, z) is an arithmetic Fuchsian group with signature $(1; e)$. We can also determine the quaternion algebra A over k associated with each Γ . We shall give the discriminant $D(A)$ of A explicitly.

THEOREM 4.1. *The complete list of all (x, y, z) such that the group Γ generated by (α, β, γ) obtained from (x, y, z) is an arithmetic Fuchsian group with signature $(1; e)$ is as follows:*

(i) $e = \infty$.

k	(x, y, z)	$D(A)$
$\mathcal{Q}$	$(\sqrt{5}, 2\sqrt{5}, 5)$	(1)
$\mathcal{Q}$	$(\sqrt{6}, 2\sqrt{3}, 3\sqrt{2})$	(1)
$\mathcal{Q}$	$(2\sqrt{2}, 2\sqrt{2}, 4)$	(1)
$\mathcal{Q}$	$(3, 3, 3)$	(1)

(ii) $e = 2$.

$\mathcal{Q}$	$(\sqrt{5}, 2\sqrt{3}, \sqrt{15})$	(2)(3)
$\mathcal{Q}$	$(\sqrt{6}, 2\sqrt{2}, 2\sqrt{3})$	(2)(3)
$\mathcal{Q}$	$(\sqrt{7}, \sqrt{7}, 3)$	(2)(7)
$\mathcal{Q}(\sqrt{5})$	$(\sqrt{2w_5+2}, \sqrt{4w_5+4}, \sqrt{6w_5+4})$	$\mathfrak{p}_2$
$\mathcal{Q}(\sqrt{5})$	$(\sqrt{3w_5+2}, \sqrt{3w_5+2}, \sqrt{4w_5+4})$	$\mathfrak{p}_2$
$\mathcal{Q}(\sqrt{5})$	$(\sqrt{3w_5+2}, \sqrt{3w_5+3}, \sqrt{3w_5+3})$	$\mathfrak{p}_2$
$\mathcal{Q}(\sqrt{2})$	$(\sqrt{w_8+3}, \sqrt{8w_8+12}, \sqrt{9w_8+13})$	$\mathfrak{p}_7 (= (3+\sqrt{2}))$
$\mathcal{Q}(\sqrt{2})$	$(\sqrt{2w_8+3}, \sqrt{3w_8+5}, \sqrt{3w_8+5})$	$\mathfrak{p}'_7 (= (3-\sqrt{2}))$
$\mathcal{Q}(\sqrt{2})$	$(\sqrt{2w_8+4}, \sqrt{2w_8+4}, \sqrt{4w_8+6})$	$\mathfrak{p}_2 (= (\sqrt{2}))$
$\mathcal{Q}(\sqrt{3})$	$(\sqrt{w_{12}+3}, \sqrt{4w_{12}+8}, \sqrt{5w_{12}+9})$	$\mathfrak{p}_3 (= (\sqrt{3}))$
$\mathcal{Q}(\sqrt{3})$	$(\sqrt{2w_{12}+4}, \sqrt{2w_{12}+4}, \sqrt{2w_{12}+4})$	$\mathfrak{p}_2$
$\mathcal{Q}(\sqrt{13})$	$(\sqrt{w_{13}+2}, \sqrt{8w_{13}+12}, \sqrt{9w_{13}+12})$	$\mathfrak{p}_2\mathfrak{p}_3\mathfrak{p}'_3$
$\mathcal{Q}(\sqrt{13})$	$(\sqrt{w_{13}+3}, \sqrt{3w_{13}+4}, \sqrt{3w_{13}+4})$	$\mathfrak{p}_2$
$\mathcal{Q}(\sqrt{17})$	$(\sqrt{w_{17}+2}, \sqrt{4w_{17}+8}, \sqrt{5w_{17}+8})$	$\mathfrak{p}'_2 (= (w'_{17}+2))$
$\mathcal{Q}(\sqrt{17})$	$(\sqrt{w_{17}+3}, \sqrt{2w_{17}+4}, \sqrt{3w_{17}+5})$	$\mathfrak{p}_2 (= (w_{17}+2))$
$\mathcal{Q}(\sqrt{21})$	$(\sqrt{w_{21}+2}, \sqrt{3w_{21}+6}, \sqrt{3w_{21}+7})$	$\mathfrak{p}_2$

$$\begin{array}{lll}
\mathbf{Q}(\sqrt{6}) & (\sqrt{w_{24}+3}, \sqrt{2w_{24}+5}, \sqrt{2w_{24}+6}) & \mathfrak{p}_2 (= (w_{24}+2)) \\
\mathbf{Q}(\sqrt{33}) & (\sqrt{w_{33}+3}, \sqrt{w_{33}+4}, \sqrt{2w_{33}+5}) & \mathfrak{p}_2 (= (w_{33}-3))
\end{array}$$

We define w_d for the discriminant d of a quadratic field $\mathbf{Q}(\sqrt{d})$ as follows:

$$(4.1) \quad w_d = \begin{cases} (1+\sqrt{d})/2 & \text{if } d \equiv 1 \pmod{4}, \\ \sqrt{d}/2 & \text{if } d \equiv 0 \pmod{4}. \end{cases}$$

$f(t)$	$d(k)$	ρ	
t^3-t^2-2t+1	49	$2\cos(\pi/7)$	$(\sqrt{\rho^2+\rho}, \sqrt{3\rho^2+2\rho-1}, \sqrt{3\rho^2+2\rho-1})$ $\mathfrak{p}_2\mathfrak{p}_7$
t^3-3t-1	81	$\rho \doteq 1.8794$	$(\sqrt{\rho^2+\rho+1}, \rho+1, \rho+1)$ (1)
t^3-t^2-3t+1	148	$\rho \doteq 2.1700$	$(\sqrt{\rho^2+\rho}, \sqrt{\rho^2+\rho}, \sqrt{\rho^2+2\rho+1})$ (1)
t^3-t^2-3t+1	148	$\rho \doteq 0.3111$	$(\sqrt{-\rho^2+\rho+4}, \sqrt{-12\rho^2+8\rho+40},$ $\sqrt{-13\rho^2+9\rho+42})$ (1)
t^3-t^2-3t+1	148	$\rho \doteq -1.4811$	$(\sqrt{\rho^2-2\rho+1}, \sqrt{\rho^2-3\rho+2}, \sqrt{\rho^2-3\rho+2})$ (1)
t^3-4t-1	229	$\rho \doteq 2.1149$	$(\sqrt{\rho+2}, \sqrt{8\rho^2+16\rho+4}, \sqrt{8\rho^2+17\rho+4})$ $\mathfrak{p}_2\mathfrak{p}'_2$
t^3-4t-1	229	$\rho \doteq -0.2541$	$(\sqrt{-\rho^2+5}, \sqrt{-3\rho^2+\rho+13},$ $\sqrt{-4\rho^2+\rho+16})$ $\mathfrak{p}_2\mathfrak{p}'_2$
t^3-4t-1	229	$\rho \doteq -1.8608$	$(\sqrt{\rho^2-2\rho}, \sqrt{\rho^2-2\rho}, \sqrt{\rho^2-2\rho+1})$ $\mathfrak{p}_2\mathfrak{p}'_2$
$t^4-t^3-3t^2+t+1$	725	$\rho \doteq -1.3556$	$(x=\sqrt{-\rho^3+2\rho^2+\rho},$ $y=z=\sqrt{-2\rho^3+5\rho^2-\rho-1})$ $\mathfrak{p}_2$
$t^4-t^3-3t^2+t+1$	725	$\rho \doteq -0.4772$	$(x=\sqrt{\rho^3-2\rho^2-2\rho+4},$ $y=z=\sqrt{9\rho^3-13\rho^2-21\rho+19})$ $\mathfrak{p}_2$
$t^4-t^3-4t^2+4t+1$	1125	$\rho \doteq -1.9562$	$(\sqrt{\rho^2-\rho}, \sqrt{\rho^2-2\rho+1},$ $\sqrt{-\rho^3+\rho^2+\rho+1})$ $\mathfrak{p}_2$

where $f(t)$ denotes the irreducible polynomial of ρ over $\mathbf{Q}$ such that $k=\mathbf{Q}(\rho)$.

(iii) $e=3$.

$$\begin{array}{lll}
\mathbf{Q} & (\sqrt{5}, 4, 2\sqrt{5}) & (3)(5) \\
\mathbf{Q} & (\sqrt{6}, \sqrt{10}, \sqrt{15}) & (2)(5) \\
\mathbf{Q} & (\sqrt{7}, 2\sqrt{2}, \sqrt{14}) & (2)(3)
\end{array}$$

Q	$(2\sqrt{2}, 2\sqrt{2}, 3)$	$(2)(3)$
$Q(\sqrt{5})$	$(\sqrt{2w_5+2}, \sqrt{6w_5+4}, \sqrt{8w_5+5})$	$\mathfrak{p}_3$
$Q(\sqrt{5})$	$(\sqrt{3w_5+2}, \sqrt{4w_5+3}, \sqrt{4w_5+3})$	$\mathfrak{p}_5$
$Q(\sqrt{2})$	$(\sqrt{2w_8+3}, \sqrt{4w_8+6}, \sqrt{4w_8+6})$	$\mathfrak{p}_3$
$Q(\sqrt{3})$	$(\sqrt{2w_{12}+4}, \sqrt{2w_{12}+4}, \sqrt{4w_{12}+7})$	$\mathfrak{p}_3$
$Q(\sqrt{13})$	$(\sqrt{2w_{13}+3}, \sqrt{2w_{13}+3}, \sqrt{3w_{13}+4})$	$\mathfrak{p}'_3 (= (w'_{13}))$
$Q(\sqrt{13})$	$(\sqrt{w_{13}+2}, \sqrt{12w_{13}+16}, \sqrt{13w_{13}+17})$	$\mathfrak{p}_3 (= (w_{13}))$
$Q(\sqrt{17})$	$(\sqrt{w_{17}+2}, \sqrt{6w_{17}+10}, \sqrt{7w_{17}+11})$	$\mathfrak{p}_2\mathfrak{p}'_2\mathfrak{p}_3$
$Q(\sqrt{21})$	$(\sqrt{w_{21}+2}, \sqrt{4w_{21}+8}, \sqrt{5w_{21}+9})$	$\mathfrak{p}_3$
$Q(\sqrt{7})$	$(\sqrt{w_{28}+3}, \sqrt{2w_{28}+6}, \sqrt{3w_{28}+8})$	$\mathfrak{p}_2\mathfrak{p}_3\mathfrak{p}'_3$

$f(t)$	$d(k)$	ρ	
t^3-t^2-2t+1	49	$2\cos(\pi/7)$	$(x=\sqrt{\rho^2+\rho}, y=z=\sqrt{4\rho^2+3\rho-2})$ (1)
t^3-3t^2+1	81	$-1/(2\cos(5\pi/9))$	(ρ, ρ, ρ) (1)

where $f(t)$ denotes the irreducible polynomial of ρ over $\mathbf{Q}$ such that $k=Q(\rho)$.

(iv) $e=4$.

$Q(\sqrt{2})$	$(\sqrt{3+\sqrt{2}}, \sqrt{20+12\sqrt{2}}, \sqrt{21+14\sqrt{2}})$	$\mathfrak{p}_2\mathfrak{p}_7\mathfrak{p}'_7$
$Q(\sqrt{2})$	$(\sqrt{4+\sqrt{2}}, \sqrt{8+4\sqrt{2}}, \sqrt{10+6\sqrt{2}})$	$\mathfrak{p}_7 (= (3-\sqrt{2}))$
$Q(\sqrt{2})$	$(\sqrt{3+2\sqrt{2}}, \sqrt{7+4\sqrt{2}}, \sqrt{7+4\sqrt{2}})$	$\mathfrak{p}_2$
$Q(\sqrt{2})$	$(\sqrt{3+2\sqrt{2}}, \sqrt{6+4\sqrt{2}}, \sqrt{9+4\sqrt{2}})$	$\mathfrak{p}_2$
$Q(\sqrt{2})$	$(\sqrt{4+2\sqrt{2}}, \sqrt{6+2\sqrt{2}}, \sqrt{8+5\sqrt{2}})$	$\mathfrak{p}_7 (= (3+\sqrt{2}))$
$Q(\sqrt{2})$	$(\sqrt{5+2\sqrt{2}}, \sqrt{5+2\sqrt{2}}, \sqrt{6+4\sqrt{2}})$	$\mathfrak{p}_2$
$Q(\sqrt{7-2\sqrt{2}})$	$d(k)=2624_{\frac{7}{2}} \quad \rho=(1+\sqrt{13+8\sqrt{2}})/2$ $(x=\sqrt{\rho+2}, y=z=\sqrt{(1+2\sqrt{2})\rho+5+2\sqrt{2}})$	$\mathfrak{p}_2 (= (\sqrt{2}))$
$Q(\sqrt{7+2\sqrt{2}})$	$d(k)=2624 \quad \rho=(1+\sqrt{2}+\sqrt{7+2\sqrt{2}})/2$ $(x, y, z)=(\rho, \rho, \rho+1)$	$\mathfrak{p}_2$
$Q(\sqrt{2}, \sqrt{3})$	$d(k)=2304 \quad \rho=(2+\sqrt{2}+\sqrt{6})/2$ $(x, y, z)=(\rho, \rho, \rho)$	$\mathfrak{p}_2$

(v) $e=5$.

$$\begin{array}{lll}
 Q(\sqrt{5}) & (\sqrt{w_5+3}, \sqrt{12w_5+8}, \sqrt{14w_5+9}) & \mathfrak{p}_5 (= (\sqrt{5})) \\
 Q(\sqrt{5}) & (\sqrt{2w_5+2}, \sqrt{6w_5+6}, \sqrt{9w_5+6}) & \mathfrak{p}_2\mathfrak{p}_3\mathfrak{p}_5 \\
 Q(\sqrt{5}) & (\sqrt{2w_5+3}, \sqrt{4w_5+4}, \sqrt{7w_5+5}) & \mathfrak{p}_5 \\
 Q(\sqrt{5}) & (\sqrt{3w_5+2}, \sqrt{4w_5+4}, \sqrt{4w_5+4}) & \mathfrak{p}_5 \\
 Q(\sqrt{5}) & (\sqrt{3w_5+3}, \sqrt{3w_5+3}, \sqrt{5w_5+5}) & \mathfrak{p}_2 \\
 Q(\sqrt{13w_5+9}) & d(k)=725 \quad \rho=(w_5+\sqrt{13w_5+9})/2 \\
 & (x=\sqrt{\rho+2}, y=z=\sqrt{(w_5+1)\rho+2w_5+2}) & \mathfrak{p}_5 \\
 Q(\sqrt{7w_5+6}) & d(k)=725 \quad \rho=(w_5+3+\sqrt{7w_5+6})/2 \\
 & (x=\sqrt{\rho}, y=z=\sqrt{(5w_5+2)\rho-2w_5+1}) & \mathfrak{p}_5 \\
 Q(\sqrt{33w_5+21}) & d(k)=1125 \quad \rho=(1+w_5+(2-w_5)\sqrt{33w_5+21})/2 \\
 & (x, y, z)=(\rho, \rho, \rho) & \mathfrak{p}_5
 \end{array}$$

(vi) $e=6$.

$$\begin{array}{lll}
 Q(\sqrt{3}) & (\sqrt{3+\sqrt{3}}, \sqrt{14+6\sqrt{3}}, \sqrt{15+8\sqrt{3}}) & \mathfrak{p}_2\mathfrak{p}_3\mathfrak{p}_{11} \\
 Q(\sqrt{3}) & (\sqrt{5+\sqrt{3}}, \sqrt{6+2\sqrt{3}}, \sqrt{9+4\sqrt{3}}) & \mathfrak{p}_2\mathfrak{p}_3\mathfrak{p}_{11}
 \end{array}$$

(vii) $e=7$.

$$\begin{array}{lll}
 k=Q(\cos(\pi/7)) & d(k)=49 \quad \rho=2\cos(\pi/7) & \\
 (x, y, z)=(\sqrt{\rho^2+1}, \sqrt{16\rho^2+12\rho-8}, \sqrt{17\rho^2+13\rho-9}) & & \mathfrak{p}_7\mathfrak{p}_{13} \\
 (\sqrt{\rho^2+\rho}, \sqrt{5\rho^2+3\rho-2}, \sqrt{5\rho^2+3\rho-2}) & & \mathfrak{p}_7\mathfrak{p}'_{13} \\
 (\sqrt{2\rho^2+\rho}, \sqrt{2\rho^2+\rho}, \sqrt{3\rho^2+\rho-1}) & & \mathfrak{p}_7\mathfrak{p}''_{13} \\
 (\sqrt{2\rho^2}, \sqrt{2\rho^2+2\rho}, \sqrt{4\rho^2+3\rho-2}) & & (1)
 \end{array}$$

(viii) $e=9$.

$$\begin{array}{lll}
 k=Q(\cos(\pi/9)) & d(k)=81 \quad \rho=2\cos(\pi/9) & \\
 (x, y, z)=(\sqrt{\rho^2+1}, \sqrt{4\rho^2+8\rho+4}, \sqrt{5\rho^2+9\rho+3}) & & \mathfrak{p}_3\mathfrak{p}_{17} \\
 (\sqrt{\rho^2+\rho+1}, \sqrt{2\rho^2+2\rho+1}, \sqrt{2\rho^2+2\rho+1}) & & \mathfrak{p}_3\mathfrak{p}'_{17} \\
 (\sqrt{\rho^2+2\rho+1}, \sqrt{\rho^2+2\rho+2}, \sqrt{\rho^2+2\rho+2}) & & \mathfrak{p}_3\mathfrak{p}''_{17}
 \end{array}$$

(ix) $e=11$.

$$\begin{array}{lll}
 k=Q(\cos(\pi/11)) & d(k)=11^4 \quad \rho=2\cos(\pi/11) & \\
 (x, y, z)=(\rho^2-1, \rho^3-2\rho, \rho^3-2\rho) & & (1)
 \end{array}$$

where we denote by $\mathfrak{p}_p$ the prime ideal of k dividing (p) for a prime number p .

We shall give the proof of Theorem 4.1 in 4.2-4.10. We have only to deal with the cases (e, n) listed in Theorem 3.6 and the case $e=\infty$. Let Γ be an arithmetic Fuchsian group with signature $(1; e)$. Then by Proposition 3.1 and (2.5) we have

$$(4.2) \quad (2\pi)^{-2n} d(k)^{3/2} \cdot \zeta_k(2) \prod_{\mathfrak{p} \mid D(A)} (n_{k/Q}(\mathfrak{p}) - 1) = d_2^{-1} (1 - 1/e),$$

where $d_2 = [\Gamma^{(1)}(A, O_1) : \Gamma^{(2)} \cdot \{\pm 1_2\}]$. In solving the simultaneous linear inequalities we used a programable electronic calculator YHP-67. In view of Theorem 3.4 the general procedure to obtain all solutions (x, y, z) over k is as follows. Let $\{w_1, \dots, w_n\}$ be a $\mathbf{Z}$ -basis of O_k . Then we have the expressions $x^2 = m_1 w_1 + \dots + m_n w_n$, $y^2 = r_1 w_1 + \dots + r_n w_n$, $z^2 = s_1 w_1 + \dots + s_n w_n$ ($m_i, r_i, s_i \in \mathbf{Z}$). From (3.10), (3.12) we have the simultaneous inequalities for $(m_1, \dots, m_n)$. For each solution x^2 by (3.10), (3.11) we have the inequalities for $(r_1, \dots, r_n)$. For each (x^2, y^2) we have the inequalities for $(s_1, \dots, s_n)$. Finally for each (x, y, z) we check the condition (3.9). We note here that x, y, z are not necessarily contained in k .

Let us consider the case $e=\infty$. Since Γ contains a parabolic element in this case, it is well-known that $k=\mathbf{Q}$, $A \cong M_2(\mathbf{Q})$. From (3.10) we see that x^2 is a rational integer such that $5 \leq x^2 \leq 9$. For each x^2 we can easily solve the inequalities for y^2, z^2 . Thus, we can obtain all solutions in the case $e=\infty$.

4.2. The case $e=2$.

Let us consider the case $e=2$. In this case by Theorem 3.6 we have $1 \leq n \leq 6$. Furthermore, from (3.10), (3.12) we have

$$(4.3) \quad 4 < x^2 \leq 4 + 2\sqrt{3}, \quad 0 < \varphi_i(x^2) < 2 \quad (2 \leq i \leq n).$$

In the case $n=1$ it is easy to obtain all solutions. Let us consider the case $n=2$. Let w_d be the same as in (4.1). Then $\{1, w_d\}$ is a $\mathbf{Z}$ -basis of the ring O_k of integers in k . Put $x^2 = a + b w_d$ ($a, b \in \mathbf{Z}$). Then from (4.3) we have $2 < b\sqrt{d} \leq 4 + 2\sqrt{3}$. Hence $d \leq (4 + 2\sqrt{3})^2 = 55.7\dots$. Thus we have $d = d(k) \leq 55$. For each d we can obtain all solutions x^2 in O_k satisfying (4.3). For each x^2 we can solve the inequalities of Theorem 3.4 for y, z . Hence we obtain all solutions in the case $n=2$.

Let us consider the case $n=3$. Since $\zeta_k(2) > 1$ and $\prod_{\mathfrak{p} \mid D(A)} (n_{k/Q}(\mathfrak{p}) - 1) \geq 1$, from (4.2) we have $d(k) < ((2\pi)^6/2)^{2/3} = 981.822\dots$. Hence we have $d(k) \leq 981$. A list of the totally real algebraic number fields k of degree 3 with small $d(k)$ can be found in K. K. Billevich [1] p. 134 and in B. N. Delone-D. K. Faddeev [2] p. 159. In view of these lists we obtain the following 25 cases:

$d(k)=49, 81, 148, 169, 229, 257, 316, 321, 361, 404, 469, 473, 564, 568,$
 $621, 697, 733, 756, 761, 785, 788, 837, 892, 940, 961.$

For each $d(k)$ listed above the defining equation for k and a $\mathbf{Z}$ -basis of O_k are given in [1], [2]. Using those data we can compute the relative degrees f_p and the ramification indexes e_p for the prime ideals $\mathfrak{p}$ of k dividing the prime numbers $p=2, 3, 5$. Thus, we can compute the $\mathfrak{p}$ -factor of the Euler product $\zeta_k(2) = \prod_{\mathfrak{p}} (1 - n_{k/\mathbf{Q}}(\mathfrak{p})^{-2})^{-1}$. It implies that the cases $d(k)=788, 837, 892, 940, 961$ are excluded because the left hand side of (4.2) is greater than $1/2$ in these cases. In each remaining case a $\mathbf{Z}$ -basis of O_k is given. Therefore following the general procedure we can obtain the solutions for x^2 and then for y, z .

Let us consider the case $n=4$. From (4.2) we have $d(k) < ((2\pi)^8/2)^{2/3} = 11383.416\ldots$. Hence we have $d(k) \leq 11383$. A list of the totally real algebraic number fields k with $d(k) \leq 11664$ is given by H.J. Godwin [4]. A list of such fields k with $d(k) \leq 8112$ (resp. 7168) is also given in [2] (resp. [1]). A $\mathbf{Z}$ -basis of O_k for each k is also given there. Using these data we can obtain all solutions. However, in order to avoid the extensive numerical computations we make the following arguments.

We distinguish two cases $2 < y^2 - x^2$ and $0 \leq y^2 - x^2 \leq 2$. First let us consider the former case. From (3.10) we have $x^2 + 2 < (x^2 - 2)/(x - 2)$. Solving this inequality numerically we have

$$(4.4) \quad 4 < x^2 < 6.4, \quad 0 < \varphi_i(x^2) < 2 \quad (2 \leq i \leq 4).$$

It follows from (4.4) that $|n_{k/\mathbf{Q}}(x^2(2 - x^2)(1 - x^2)^2)| < 12.83\ldots$. Hence we have

$$(4.5) \quad -12 \leq n_{k/\mathbf{Q}}(x^2(2 - x^2)(1 - x^2)^2) \leq -1.$$

Let $f(t) = t^4 + a_3t^3 + a_2t^2 + a_1t + a_0$ ($a_i \in \mathbf{Z}$) be the irreducible polynomial of x^2 over $\mathbf{Q}$. Let $b_i = f(i)$ ($0 \leq i \leq 2$). Then from (4.4), (4.5) we have

$$(4.6) \quad b_0 \geq 1, \quad b_2 \leq -1, \quad -12 \leq b_0b_1^2b_2 \leq -1.$$

We can determine easily all triples (b_0, b_1, b_2) satisfying (4.6). Since $a_3 = -\text{tr}_{k/\mathbf{Q}}(x^2)$, from (4.4) we have

$$(4.7) \quad -12 \leq a_3 \leq -5.$$

Using the expressions: $a_0 = b_0$, $a_1 = (-b_2 + 4b_1 - 3b_0 + 4a_3 + 12)/2$, $a_2 = (b_2 - 2b_1 + b_0 - 6a_3 - 14)/2$, we can obtain all (a_3, a_2, a_1, a_0) such that x^2 satisfies (4.4), which are as follows:

a_3	a_2	a_1	a_0	$d(f)$
-7	13	-7	1	725 ($=5^2 \cdot 29$)
-8	18	-13	1	725
-8	14	-7	1	1125 ($=3^2 \cdot 5^3$)
-9	20	-14	3	1957 ($=19 \cdot 103$)

-10	27	-26	7	2624 ($=2^4 \cdot 41$)
-9	22	-18	3	3981 ($=3 \cdot 1327$)
-8	15	-8	1	4752 ($=2^4 \cdot 3^3 \cdot 11$)
-8	16	-9	1	8069 (prime)
-9	19	-12	2	11324 ($=2^2 \cdot 19 \cdot 149$)
-9	21	-15	1	14197 (prime)
-9	19	-11	1	36677 (prime),

where we denote by $d(f)$ the discriminant of $f(t)$. It is known that 725 is the smallest discriminant of the totally real algebraic number fields of degree 4. Since $d(f)=d(k)m^2$ ($m \in \mathbb{Z}$), in view of the list in Godwin [4] we see that $d(f)=d(k)$ and $O_k=\mathbb{Z}[x^2]$ in each case listed above. Since $n=4$ is even, by (2.1) and the Hasse's principle we see that the number of the prime ideals of k dividing $D(A)$ is odd. In particular, $D(A) \neq (1)$. In the cases: $d(f)=8069, 11324, 14197, 36677$ we can see easily that the left hand side of (4.2) is greater than $1/2$. Hence these cases are excluded. In the remaining cases we can obtain all solutions following the general procedure.

Now let us consider the second case $0 \leq y^2 - x^2 \leq 2$. Let $a=y^2 - x^2$. Then from (3.10), (3.12) we have

$$(4.8) \quad 0 \leq a < 2, \quad -2 < \varphi_i(a) < 2 \quad (2 \leq i \leq 4).$$

We need the following lemma (cf. Pólya-Szegő [12] p. 145).

LEMMA 4.2. *Let a be a totally real algebraic integer such that all conjugates $\varphi_i(a)$ of a satisfy the inequalities $-2 \leq \varphi_i(a) \leq 2$ ($1 \leq i \leq n$). Then $a=2\cos(2\pi r)$ ($r \in \mathbb{Q}$).*

From this lemma we have $a=2\cos(2\pi r)$ ($r \in \mathbb{Q}$). By (3.10), (3.11) we have

$$(4.9) \quad \begin{cases} (8-a+\sqrt{a^2+32})/2 \leq x^2 < 4+2\sqrt{3}, \\ 0 < \varphi_i(x^2) \leq (8-\varphi_i(a)-\sqrt{\varphi_i(a)^2+32})/2 \quad (2 \leq i \leq n). \end{cases}$$

Moreover,

$$(4.10) \quad \text{If } \varphi_i(a) < 0, \text{ then } -\varphi_i(a) < \varphi_i(x^2) \quad (2 \leq i \leq n).$$

Since a is contained in k , we see that $[Q(a):Q]=1, 2, 4$. Assume that $a \in Q$. Then $a=0$ or 1 . In these cases from (4.9) we see that $0 < \varphi_i(x^2) \leq 4-2\sqrt{2}$ or $(7-\sqrt{33})/2$. Hence we have $|n_{k/Q}(x^2(1-x^2))| < 1$, which is a contradiction.

Let us consider the case $[Q(a):Q]=2$. Then we see that $a=2\cos(\pi/4), 2\cos(\pi/5), 2\cos(2\pi/5)$ or $2\cos(\pi/6)$. Let $v_1=\text{tr}_{k/Q}(x^2)$, $v_0=n_{k/Q}(x^2)$. Then by (4.9) and (4.10) we obtain the inequalities for v_0, v_1 and their Q -conjugates v'_0, v'_1 . Solving these inequalities for each case, we see that there exist no solutions in each case.

Let us consider the case $[Q(a):Q]=4$. We have $k=Q(a)=Q(\cos(\pi/8))$, $Q(\cos(\pi/10))$, $Q(\cos(\pi/12))$ or $Q(\cos(\pi/15))$. Since we know that $\zeta_k(2)=2^{-5}3^{-1}5(2\pi)^8d(k)^{-3/2}$ for $k=Q(\cos(\pi/8))$ (cf. [18] p. 208), from (4.2) we have $\prod_{p|D(A)} (n_{k/Q}(p)-1)d_2=2^43/5$, which is not an integer. This is a contradiction.

In order to deal with the remaining cases we need the following

LEMMA 4.3. *Let $\rho=2\cos(2\pi/m)$ ($m \in \mathbb{Z}$) and $k=Q(\rho)$. Then $\{1, \rho, \rho^2, \dots, \rho^{d-1}\}$ is a $\mathbb{Z}$ -basis of the ring O_k of integers in k , where $d=[k:Q]$.*

The proof of this lemma is referred to Liang [9]. Let $\rho_r=2\cos(\pi/r)$ for each case $k=Q(\cos(\pi/r))$, $r=10, 12$ or 15 . By (3.10) we have the inequality $4(x^2-2)/(x^2-4) < x^2+2$ in this case. Solving this inequality numerically, we have

$$(4.10) \quad 6 < x^2 < 4+2\sqrt{3}, \quad 0 < \varphi_i(x^2) < 2 \quad (2 \leq i \leq 4).$$

By Lemma 4.3 we have the expression $x^2 = \sum_{0 \leq i \leq 3} m_i \rho_r^i$ ($m_i \in \mathbb{Z}$). Solving the inequalities for (m_0, m_1, m_2, m_3) given by (4.10) numerically, we see that there exist no solutions for x^2 .

Let us consider the case $n=5$. Solving the inequality numerically $1 \leq |n_{k/Q}(x^2(2-x^2)(1-x^2)^2)| \leq x^2(x^2-2)(x^2-1)^2/4^4$, we have

$$(4.11) \quad 5.06 < x^2 < 4+2\sqrt{3}, \quad 0 < \varphi_i(x^2) < 2 \quad (2 \leq i \leq 5).$$

We shall show that if $0 \leq y^2 - x^2 \leq 2$ or $0 \leq z^2 - y^2 \leq 2$, then $k=Q(\cos(\pi/11))$. Assume that $0 \leq y^2 - x^2 \leq 2$. Then by (3.12) and Lemma 4.2 we have $y^2 - x^2 = 2\cos(2\pi r)$, $r \in Q$. Since $n=5$, we have $y^2 - x^2 = 0, 1$ or $2\cos(2\pi s/11)$. If $y^2 - x^2 = 0$ or 1 , then by (4.9) we have $|n_{k/Q}(x^2(1-x^2))| < 1$, which is a contradiction. Assume that $0 \leq z^2 - y^2 \leq 2$. Then we have $z^2 - y^2 = 0, 1$ or $2\cos(2\pi s/11)$. If $z^2 - y^2 = 0$ or 1 , then by the fact that the function $(x^2-2)/(x-2)$ is monotone-decreasing on $\sqrt{5.06} < x < 3$, from (3.10), (3.11) we have

$$y^2 < 12.268, \quad 0 < \varphi_i(y^2) < 4-2\sqrt{2} \quad (\text{resp. } (7-\sqrt{33})/2) \quad (2 \leq i \leq 5).$$

It follows that $|n_{k/Q}(y^2(1-y^2))| < 1$, which is a contradiction. Hence we see that $k=Q(\cos(\pi/11))$. Since $\zeta_k(2)=2^{-3} \cdot 3^{-1} \cdot 5 \cdot 11^{-1} (2\pi)^{10} d(k)^{-3/2}$ for $k=Q(\cos(\pi/11))$ (cf. [18] p. 208), we have $\prod_{p|D(A)} (n_{k/Q}(p)-1)d_2=2^2 \cdot 3 \cdot 11/5$, which is not an integer.

This is a contradiction.

Now we must consider the case:

$$(4.12) \quad x^2+2 < y^2, \quad y^2+2 < z^2.$$

From (3.10) and the second inequality of (4.12) we have

$$(4.13) \quad y^2 < x^2(1+\sqrt{x^2-3})/(x^2-4).$$

Combining the first inequality of (4.12) with (4.13), we have

$$(4.14) \quad 5.06 < x^2 < 6.071, \quad 0 < \varphi_i(x^2) < 2 \quad (2 \leq i \leq 5).$$

From (4.14) we have $|n_{k/Q}(x^2(2-x^2)(1-x^2)^2)| < 2.48 \dots$. Hence we have $n_{k/Q}(1-x^2) = \pm 1$, $n_{k/Q}(x^2(2-x^2)) = -1$ or -2 . Since $x^2 + (2-x^2) = 2$, $n_{k/Q}(x^2)$ is divisible by 2 if and only if $n_{k/Q}(2-x^2)$ is so. Therefore we have

$$(4.15) \quad n_{k/Q}(-x^2) = -1, \quad n_{k/Q}(2-x^2) = -1, \quad n_{k/Q}(1-x^2) = \pm 1.$$

Let $f(t) = t^5 + a_4t^4 + a_3t^3 + a_2t^2 + a_1t + a_0$ ($a_i \in \mathbf{Z}$) be the irreducible polynomial of x^2 over $\mathbf{Q}$. Then from (4.15) we have $f(0) = -1$, $f(2) = -1$, $f(1) = \pm 1$.

We distinguish two cases: $f(1) = 1$ and -1 . Let us consider first the case $f(1) = -1$. In this case we have the expression: $f(t) = t(t-1)(t-2)(t^2 + c_1t + c_0) - 1$ ($c_i \in \mathbf{Z}$). Since $\text{tr}_{k/Q}(x^2) = 3 - c_1$, by (4.14) we have

$$(4.16) \quad -11 \leq c_1 \leq -3.$$

From (4.14) we have $f(5.06) < 0 < f(6.071)$. This gives the inequalities for (c_0, c_1) . Solving these inequalities numerically, we have a finite set of solutions for (c_0, c_1) . We check the condition (4.14) for each case (c_0, c_1) . Hence we have only one case: $f(t) = t^5 - 10t^4 + 29t^3 - 32t^2 + 12t - 1$, $d(f) = 24217$. However, solving the inequalities (3.10), (3.11) for y^2 in this case, we see that there exist no solutions.

For the case $f(1) = 1$ by the same way as in the case $f(1) = -1$ we see that there exist no solutions.

Let us consider the case $n = 6$. From the inequality:

$$1 \leq |n_{k/Q}(x^2(2-x^2)(1-x^2)^2)| < -x^2(2-x^2)(1-x^2)^2/4^5,$$

we have

$$(4.17) \quad 6.7 < x^2 < 4 + 2\sqrt{3}, \quad 0 < \varphi_i(x^2) < 2 \quad (2 \leq i \leq 6).$$

Let $a = y^2 - x^2$. Then from (3.10) we have

$$0 \leq a \leq (x^2 - 2)/(x - 2) - x^2.$$

Since the function $(x^2 - 2)/(x - 2) - x^2$ is monotone-decreasing on $\sqrt{6.7} \leq x$, we have

$$0 \leq a \leq 1.2873, \quad -2 < \varphi_i(a) < 2 \quad (2 \leq i \leq 6).$$

By Lemma 4.2 we have $a = 2\cos(2\pi s/r)$, where $1 \leq r$, $s \in \mathbf{Z}$ such that $\varphi(r) = 2, 4, 6$ or 12 and $(r, s) = 1$ and $0 < s/r \leq 1/4$.

Assume that $a = 0$ or 1 . Then by (4.9) we have $|n_{k/Q}(x^2(1-x^2))| < 1$, which is a contradiction. There remain the cases $\varphi(r) = 4, 6$ or 12 . In these cases for each pair (r, s) we can show that $|n_{k/Q}(x^2)| < 1$ or $|n_{k/Q}(y^2)| < 1$. This is a contradiction. We have finished the case $e = 2$.

4.3. The case $e = 3$.

Let us consider the case $e = 3$. In this case we have $c_e = 1$. By Theorem 3.6 we have $1 \leq n \leq 4$. From (3.10) we have $x^2 \leq (x^2 - 1)/(x - 2)$. Solving this inequality numerically, we have

$$(4.18) \quad 4 < x^2 < 8.291, \quad 0 < \varphi_i(x^2) < 1 \quad (2 \leq i \leq n).$$

By the inequalities: $1 \leq |n_{k/Q}(x^2(1-x^2))| < 8.291 \cdot 7.291/4^{n-1}$, we have $4^{n-1} < 60.449$. Hence we see that $n=1, 2$ or 3 . In the cases $n=1, 2$ we can obtain easily all solutions.

Now let us consider the case $n=3$. We distinguish two cases: $3 < y^2 - x^2$ and $0 \leq y^2 - x^2 \leq 3$. Let us consider the first case $3 < y^2 - x^2$. By (3.10) we have $x^2 + 3 < (x^2 - 1)/(x - 2)$. Solving this inequality, we have $x^2 < 6.6947$. On the other hand, since $1 \leq |n_{k/Q}(x^2(1-x^2))| < x^2(x^2 - 1)/4^2$, we have

$$(4.19) \quad (1 + \sqrt{65})/2 < x^2 < 6.6947, \quad 0 < \varphi_i(x^2) < 1 \quad (2 \leq i \leq 3).$$

From (4.19) we see easily that

$$(4.20) \quad n_{k/Q}(x^2(1-x^2)) = -1, -2.$$

Let $f(t) = t^3 + a_2t + a_1t + a_0$ ($a_i \in \mathbf{Z}$) be the irreducible polynomial of x^2 over $\mathbf{Q}$. By (4.19) we have $-8 \leq a_2 \leq -5$. By (4.20) we have $f(0) \cdot f(1) = 1$ or 2 . From these relations we obtain a finite set of solutions for (a_0, a_1, a_2) . Checking the condition (4.19) for each (a_0, a_1, a_2) , we obtain $f(t) = t^3 - 6t^2 + 5t - 1$, $d(f) = d(k) = 49$. For this x^2 we have a solution such that $y = z$.

Let us consider the case $0 \leq y^2 - x^2 \leq 3$. Let $a = y^2 - x^2$. Then from (3.12) we have

$$(4.21) \quad -1 \leq a - 1 \leq 2, \quad -2 < \varphi_i(a - 1) < 0 \quad (2 \leq i \leq 3).$$

Since $[Q(a) : \mathbf{Q}] = 1$ or 3 , from (4.21) we have $a = 0, 1 + 2\cos(2\pi/7)$ or $1 + 2\cos(\pi/9)$. In the case $a = 0$ by (3.10), (3.11), (4.18) we have

$$(4.22) \quad 4 + 2\sqrt{3} \leq x^2 < 8.291, \quad 0 < \varphi_i(x^2) \leq 4 - 2\sqrt{3} \quad (2 \leq i \leq 3).$$

From this we obtain a solution such that $x = y = z$, $d(k) = 81$. For two other cases we see that there exist no solutions.

4.4. The case $e=4$.

Let us consider the case $e=4$. In this case we have $c_e = 2 - \sqrt{2}$. By Theorem 3.4 (i) we see that k contains $k_0 = \mathbf{Q}(\sqrt{2})$. From (3.10), (3.12) we have

$$(4.23) \quad 4 < x^2 < 9, \quad 0 < \varphi_i(x^2) < \varphi_i(2 - \sqrt{2}) \quad (2 \leq i \leq n).$$

Let $u = 1 + \sqrt{2}$. Since $0 < \varphi_i(ux^2(\sqrt{2} - ux^2)) < 1/2$, we have

$$1 \leq |n_{k/Q}(ux^2(\sqrt{2} - ux^2)(\sqrt{2} - ux^2 - 1)^2)| < 9u(9u - \sqrt{2})(9\sqrt{2}u - 1)^2/8^{n-1}.$$

Hence $8^{n-1} < 390064.58 \dots$. It follows that $n=2, 4$ or 6 . Since $x^2 \leq y^2 \leq (x^2 - 2 + \sqrt{2})/(x - 2)$, we have

$$(4.24) \quad 4 < x^2 < 8.596, \quad 0 < \varphi_i(x^2) < \varphi_i(2 - \sqrt{2}) \quad (2 \leq i \leq n).$$

In the case $n=2$ we have $k = k_0$ and it is easy to obtain all solutions.

Let us consider the case $n=4$. Then k is a quadratic extension of k_0 . Let

$a_0 = n_{k/k_0}(x^2)$, $a_1 = \text{tr}_{k/k_0}(x^2)$. From (4.24) we have the inequalities for a_0 , a_1 and their $\mathbf{Q}$ -conjugates. We obtain 11 cases for (a_0, a_1) . For each case we can calculate $d(k)$ and obtain an O_{k_0} -basis $\{1, \rho\}$ of O_k . Using the expressions $y^2 = b_0 + b_1\rho$, $z^2 = c_0 + c_1\rho$ ($b_i, c_i \in O_{k_0}$), we obtain the inequalities for b_i, c_i and their $\mathbf{Q}$ -conjugates. Solving these inequalities we obtain three solutions for (x, y, z) .

Let us consider the case $n=6$. Let $g(t) = t^3 + b_2t^2 + b_1t + b_0$ ($b_i \in O_{k_0}$) be the irreducible polynomial of ux^2 over k_0 . By (4.24) we have

$$(4.25) \quad 4u < ux^2 < 8.596u, \quad 0 < \pm \varphi_i(ux^2) < \sqrt{2} \quad (2 \leq i \leq 6),$$

where the sign $\pm$ is determined according to $\varphi_i(\sqrt{2}) = \pm\sqrt{2}$. From (4.25) we have inequalities for b_i and their $\mathbf{Q}$ -conjugates. For each solution for (b_i) we check the condition (4.25) and we see that there exist no solutions.

4.5. The case $e=5$.

Let us consider the case $e=5$. In this case from Theorem 3.6 we see that $n=2$ or 4. For the case $n=2$ we have $k = \mathbf{Q}(\sqrt{5})$ and we obtain easily all solutions. Let us consider the case $n=4$. Then k is a quadratic extension of $k_0 = \mathbf{Q}(\sqrt{5})$. From (3.10) we have the inequality $x^3 - 3x^2 + \frac{3-\sqrt{5}}{2} \leq 0$. Solving this numerically, we have

$$(4.26) \quad 4 < x^2 < 8.740.$$

Let $u = (3 + \sqrt{5})/2$. Then we have

$$(4.27) \quad 10.472 < ux^2 < 22.882, \quad 0 < \varphi_i(ux^2) < 1 \quad (2 \leq i \leq 4).$$

It implies that $|n_{k/\mathbf{Q}}(ux^2(1-ux^2))| < 7.824$. Calculating the relative degree f_p of prime numbers $p=2, 3, 5, 7$ over $k_0/\mathbf{Q}$, we have

$$(4.28) \quad n_{k/\mathbf{Q}}(ux^2(1-ux^2)) = -1, -4, -5.$$

Let $b_0 = n_{k/k_0}(ux^2)$, $b_1 = \text{tr}_{k/k_0}(ux^2)$. Then from (4.27), (4.28) we have inequalities for b_i and their $\mathbf{Q}$ -conjugates. We obtain 5 solutions for (b_i) . For each (b_i) we calculate y^2, z^2 and we obtain three solutions.

4.6. The case $e=6$.

Let us consider the case $e=6$. In this case we see that $c_e = 2 - \sqrt{3}$ and that k contains $k_0 = \mathbf{Q}(\sqrt{3})$, $n=2, 4$ or 6. Let $u = 2 + \sqrt{3}$. Then we have

$$(4.29) \quad 4u < ux^2 < 9u, \quad 0 < \varphi_i(ux^2) < 1 \quad (2 \leq i \leq n).$$

In the case $n=2$ it is easy to obtain all solutions. Let us consider the case $n=4$. Let $g(t) = t^2 + b_1t + b_0$ ($b_i \in O_{k_0}$) be the irreducible polynomial of ux^2 over k_0 . Solving the inequalities for b_i and their $\mathbf{Q}$ -conjugates derived from (4.29), we obtain three solutions for (b_0, b_1) . For these (b_0, b_1) we have $d(k) = 4752, 27792, 39744$. On the other hand, since $n=4$ is even, we have $D(A) \neq (1)$. Hence we

have $\zeta_k(2) \prod_{p|D(A)} (n_{k/Q}(p)-1) > 4/3$. It implies that $d(k) < 13209.28$. Therefore there remains only the case $k = Q(\sqrt{15+8\sqrt{3}})$, $d(k) = 4752$. However, by straightforward calculations we see that there exist no solutions. Let us consider the case $n=6$. Let $b = uy^2 - ux^2$. Then by (3.10), (3.12) we have

$$(4.30) \quad 0 \leq b, \quad -1 < \varphi_i(b) < 1 \quad (2 \leq i \leq 6).$$

We shall show that $x=y$. Assume that $b \neq 0$. From the inequalities: $1 \leq |n_{k/Q}(b^2(1-b^2))| < b^2(b^2-1)/4^5$, we have $5.701 < b$. Since $y^2 = x^2 + b/u$, we have $x^2 + 1.5276 < (x^2 - 2 + \sqrt{3})/(x-2)$. Solving this inequality, we have $x^2 < 7.893$. This implies that $|n_{k/Q}(ux^2(1-ux^2))| < 1$. This is a contradiction. Therefore we have shown that $x=y$. From (3.10) we have $x^3 - 3x^2 + 2 - \sqrt{3} \leq 0$. Solving this numerically, we have

$$(4.31) \quad ux^2 < 8.819u.$$

On the other hand, from the inequalities: $1 \leq |n_{k/Q}(ux^2(1-ux^2))| < ux^2(ux^2-1)/4^5$, we have

$$(4.32) \quad 8.709u < ux^2.$$

If $\varphi_i(ux^2(1-ux^2)) < 0.243$ for some i , then we have $|n_{k/Q}(ux^2(1-ux^2))| < 1$. This is a contradiction. Hence we have

$$(4.33) \quad 32.502 < ux^2 < 32.913, \quad 0.4 < \varphi_i(ux^2) < 0.6 \quad (2 \leq i \leq 6).$$

Let $c = \text{tr}_{k/k_0}(ux^2)$. From (4.33) we have inequalities for c and its Q -conjugates. We see easily that there exist no solutions for c in O_{k_0} .

4.7. The cases $e=7, 9$.

Let us consider the case $e=7$. By Theorem 3.6 we have $n=3$ or 6. Let $\rho = 2\cos(\pi/7)$, $k_0 = Q(\rho)$. If $n=3$, then we have $k=k_0$. Using a Z -basis $\{1, \rho, \rho^2\}$, we have inequalities for x^2 , $\varphi_i(x^2)$. We obtain four solutions for x^2 . For each x we can obtain a solution for (x, y, z) .

Let us consider the case $n=6$. In this case k is a quadratic extension of k_0 . Let $u = \rho^2 + \rho$. Then we have

$$(4.34) \quad 4u < ux^2 < 9u, \quad 0 < \varphi_i(ux^2) < 1 \quad (2 \leq i \leq 6).$$

From the inequalities: $1 \leq |n_{k/Q}(ux^2(1-ux^2))| < ux^2(ux^2-1)/4^5$ we have

$$(4.35) \quad (1 + \sqrt{4097})/2 \leq ux^2 < 9u (= 45.440\cdots).$$

It follows that $|n_{k/Q}(ux^2(1-ux^2))| < 1.97$. Hence we have

$$(4.36) \quad n_{k/Q}(ux^2) = 1, \quad n_{k/Q}(1-ux^2) = -1.$$

We put $b_0 = n_{k/k_0}(ux^2)$, $b_1 = \text{tr}_{k/k_0}(ux^2)$. Using the expressions $b_i = \sum_{0 \leq j \leq 2} b_{ij} \rho^j$

$(b_{ij} \in \mathbf{Z})$, by (4.34), (4.35), (4.36) we have inequalities for b_{ij} . For each solution (b_i) we check the condition and we see that there exist no solutions for ux^2 .

For the case $e=9$ in the same way as in the case $e=7$ we can obtain all solutions.

4.8. The cases $e=8, 15$.

Let us consider the case $e=8$. Let $\rho=2\cos(\pi/8)$ and $k_0=\mathbf{Q}(\rho)$. By Theorem 3.6 we see that $n=4, 8$. If $n=4$, then $k=k_0$ and it is known that $\zeta_{k_0}(2)=2^3 \cdot 3^{-1} \cdot 5 \cdot \pi^8 d(k_0)^{-3/2}$ (cf. [18] p. 208). Hence we have $\prod_{p|D(A)} (n_{k/\mathbf{Q}}(p)-1)d_2=2^2 \cdot 3 \cdot 7/5$.

This is a contradiction because it is not an integer. Let us consider the case $n=8$. Then k is a quadratic extension of k_0 . Since $n=8$ is even, we have $D(A) \neq (1)$. Hence we have $\zeta_k(2) \prod_{p|D(A)} (n_{k/\mathbf{Q}}(p)-1) > 4/3$. By (4.2) we have

$$(4.37) \quad d(k) < (2^{11} \cdot 3 \cdot 7 \pi^{16})^{2/3}.$$

Since $[k:k_0]=2$, by a theorem of the algebraic number theory we have

$$(4.38) \quad d(k) = d(k_0)^2 n_{k_0/\mathbf{Q}}(D(k/k_0)),$$

where $D(k/k_0)$ is the relative discriminant of the extension k/k_0 . Since $d(k_0)=2^{11}$, by (4.37) we have

$$(4.39) \quad n_{k_0/\mathbf{Q}}(D(k/k_0)) \leq 58.$$

Considering the relative degree f_p for $p=2, 3, \dots, 57$ over $k_0/\mathbf{Q}$, we have

$$(4.40) \quad n_{k_0/\mathbf{Q}}(D(k/k_0)) = 2^m \cdot q \quad (0 \leq m \leq 5, q=1, 17, 31, 47, 49).$$

Now we have the expression $k=k_0(\sqrt{\mu})$, where μ is a totally positive algebraic integer in k_0 . Note that the class number of k_0 is 1 and that every totally positive unit of k_0 is a square of some unit of k_0 . We obtain six cases for μ satisfying (4.40). Calculating $n_{k_0/\mathbf{Q}}(D(k/k_0))$ explicitly for each case μ , we obtain $n_{k_0/\mathbf{Q}}(D(k/k_0))=2^9, 2^6 \cdot 17, 2^9 \cdot 17, 2^8 \cdot 31, 2^8 \cdot 47, 2^4 \cdot 49$, which contradicts (4.40).

For the case $e=15$ similarly to the case $e=8$ we see that there exist no solutions.

4.9. The cases $e=10, 12$.

Let us consider the cases $e=10, 12$. Let $\rho=2\cos(\pi/e)$ and $k_0=\mathbf{Q}(\rho)$ for each case. By Theorem 3.6 we see that $n=4$ (and 8 for $e=12$). Since we know that $\zeta_{k_0}(2)=2^5 \cdot 3^{-1} \pi^8 d(k_0)^{-3/2}, 2^4 \cdot \pi^8 d(k_0)^{-3/2}$ for $e=10, 12$ respectively (cf. [18] p. 208), we see that $\prod_{p|D(A)} (n_{k_0/\mathbf{Q}}(p)-1)d_2$ is not an integer. Therefore only the case: $e=12, n=8$ remains. Now let us consider this case. Put $u=1/(2-\rho)$. Then u is a unit of k_0 . By (3.10), (3.12) we have

$$(4.41) \quad 4u < ux^2 < 9u (=132.06\cdots), \quad 0 < \varphi_i(ux^2) < 1 \quad (2 \leq i \leq 8).$$

From the inequalities: $1 \leq |n_{k/\mathbf{Q}}(ux^2(1-ux^2))| \leq ux^2(ux^2-1)/4^7$, we have

$$(4.42) \quad (1 + \sqrt{65537})/2 \leq ux^2.$$

Hence we have

$$(4.43) \quad 8.757 < x^2 < 9$$

Put $b = uy^2 - ux^2$. Since the function $(t^2 - 2 + \rho)/(t - 2) - t^2$ is monotone-decreasing on $2 < t$, by (3.10), (3.12) we have

$$(4.44) \quad 0 \leq b < 4.4201, \quad -1 < \varphi_i(b) < 1 \quad (2 \leq i \leq 8).$$

Since b is an algebraic integer of k , from (4.44) we have $n_{k/Q}(b^2(1-b^2)) = 0$. Hence $b = 0$. This implies that $x = y$. By (3.10) we have $z = (x^2 - \sqrt{x^4 - 8x^2 + 8 - 4\rho})/2$. Since the function $t - \sqrt{t^2 - 8t + 8 - 4\rho}$ is monotone-decreasing on $4 \leq t$, by (4.43) we have

$$(4.45) \quad z < 3.065.$$

We put $c = uz^2 - ux^2$. Then by (3.12), (4.45) we have

$$(4.46) \quad 0 \leq c < 9.351, \quad -1 < \varphi_i(c) < 1 \quad (2 \leq i \leq 8).$$

It follows that $0 \leq |n_{k/Q}(c^2(1-c^2))| < 1$. Since c is an algebraic integer, we have $c = 0$. Hence we have $x = y = z$. By (3.10) we have $x^3 - 3x^2 + 2 - \rho = 0$. We can obtain the solution $x = 1 + 2\cos(\pi/36)$. Since $[Q(x) : k_0] = 3$, we see that $k = k_0(x^2)$ is a cubic extension of k_0 . This contradicts the fact $[k : k_0] = 2$.

4.10. The remaining cases.

Let us discuss the remaining cases which are as follows by Theorem 3.6: $e = 11, 13, 14, 16, 17, 18, 19, 20, 21, 24, 25, 27, 30, 33$.

We put $\rho_e = 2\cos(\pi/e)$ for each case e . Let us consider first the case $e = 11$. Then we have $k = Q(\rho_{11})$. By Lemma 4.3 we have the expression $x^2 = \sum_{0 \leq i \leq 4} a_i \rho_{11}^i$ ($a_i \in \mathbb{Z}$). Solving the inequalities for a_i given by (3.10), (3.12), we obtain a solution for (x, y, z) .

Let us consider the case $e = 13$. Then $k = Q(\rho_{13})$. Let $\eta = \rho_{13} + 2\cos(5\pi/13)$ and $k_1 = Q(\eta)$. Then we see that $[k_1 : Q] = 3$, $[k : k_1] = 2$. It is easy to see that $\{1, \eta, \eta^2\}$ is a $\mathbb{Z}$ -basis of O_{k_1} and that $\{1, \rho_{13}\}$ is a O_{k_1} -basis of O_k . Using the expression $x^2 = a_0 + a_1 \rho_{13}$ ($a_i \in O_{k_1}$), we have inequalities for a_i from (3.10), (3.12). We solve these inequalities to see that there exist no solutions for a_i .

For the cases $e = 18, 21$ in the similar way to the case $e = 13$ we see that there exist no solutions.

Let us consider the case $e = 14$. In this case we see that $k = Q(\rho_{14})$, $[k : Q] = 6$, $d(k) = 2^6 \cdot 7^5$. Since $n = 6$ is even, we have $D(A) \neq (1)$. By using the fact that the minimum of $n_{k/Q}(\mathfrak{p})$ for all prime ideals $\mathfrak{p}$ of k is 7 we have

$$\zeta_k(2) \prod_{\mathfrak{p} | D(A)} (n_{k/Q}(\mathfrak{p}) - 1) > 7^2/8.$$

Hence we have

$$(2\pi)^{-12}d(k)^{3/2}\zeta_k(2)\prod_{p|D(A)}(n_{k/Q}(p)-1)>1.804>1-1/14.$$

This contradicts (4.2).

For each remaining case e we have $k=Q(\rho_e)$. We can calculate $d(k)$ explicitly and we see that $(2\pi)^{-2n}d(k)^{3/2}>1-1/e$ which contradicts (4.2). We have finished the proof of Theorem 4.1.

4.11. For each triple (x, y, z) listed in Theorem 4.1 we can obtain a triple (α, β, γ) determined by (3.8). This is unique up to $GL_2(\mathbf{R})$ -conjugation but not $SL_2(\mathbf{R})$ -conjugation. We have another triple $(g_0^{-1}\alpha g_0, g_0^{-1}\beta g_0, g_0^{-1}\gamma g_0)$ satisfying (3.8), where we denote $g_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. These are complete solutions for (3.8) up to $SL_2(\mathbf{R})$ -conjugation. Let Γ be the Fuchsian group generated by $\{\alpha, \beta\}$ and let A be the quaternion algebra associated with Γ . For a fixed triple (x, y, z) any Fuchsian group derived from (x, y, z) is $SL_2(\mathbf{R})$ -conjugate to Γ or $g_0^{-1}\Gamma g_0$. It depends on the case whether these two groups are $SL_2(\mathbf{R})$ -conjugate or not.

For a fixed e different triples may correspond to the same Γ . Now we shall show that each Γ derived from each triple (x, y, z) listed in Theorem 4.1 is pairwise $GL_2(\mathbf{R})$ -inconjugate. Let (x', y', z') be another triple for the fixed e . Let Γ' be the Fuchsian group derived from it and A'/k' be the quaternion algebra associated with Γ' . Suppose that $\Gamma' = g^{-1}\Gamma g$ for $g \in GL_2(\mathbf{R})$. By a result in [17] we see that $Q(\text{tr}(\gamma)|\gamma \in \Gamma) = Q(x, y, z)$. It follows that $k=k'$, $D(A)=D(A')$, $Q(x, y, z)=Q(x', y', z')$. However, in view of the data in Theorem 4.1 we see that there exist no such triples (x, y, z) and (x', y', z') .

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